Setting up TSMCN16 PDK

Create your tsmcn16 Cad start directory:
mkdir tsmcn16
cd tsmcn16

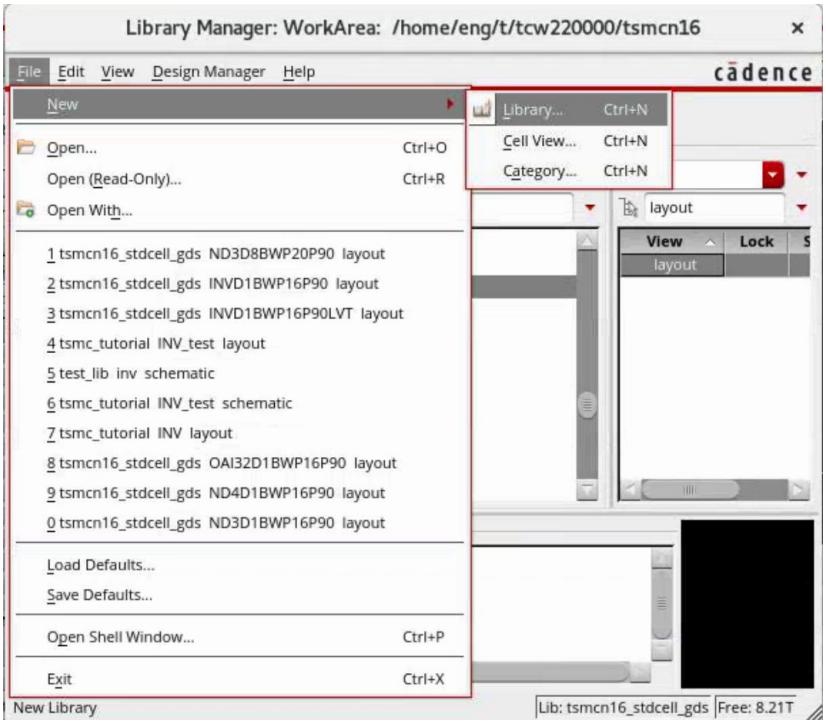
Run the setup script, only needs to be done once for setup, skip otherwise:
/proj/cad/library/imec/tsmc16_adfp/util/D-AFP-02-03-001_2_X_executable_package/Executable_Package/Collaterals/Tech/iPDK/N16ADFP_iPDK/setup_tsmcn16_script.sh

Source tsmcn16 profile:
. /proj/cad/startup/profile.tsmcn16

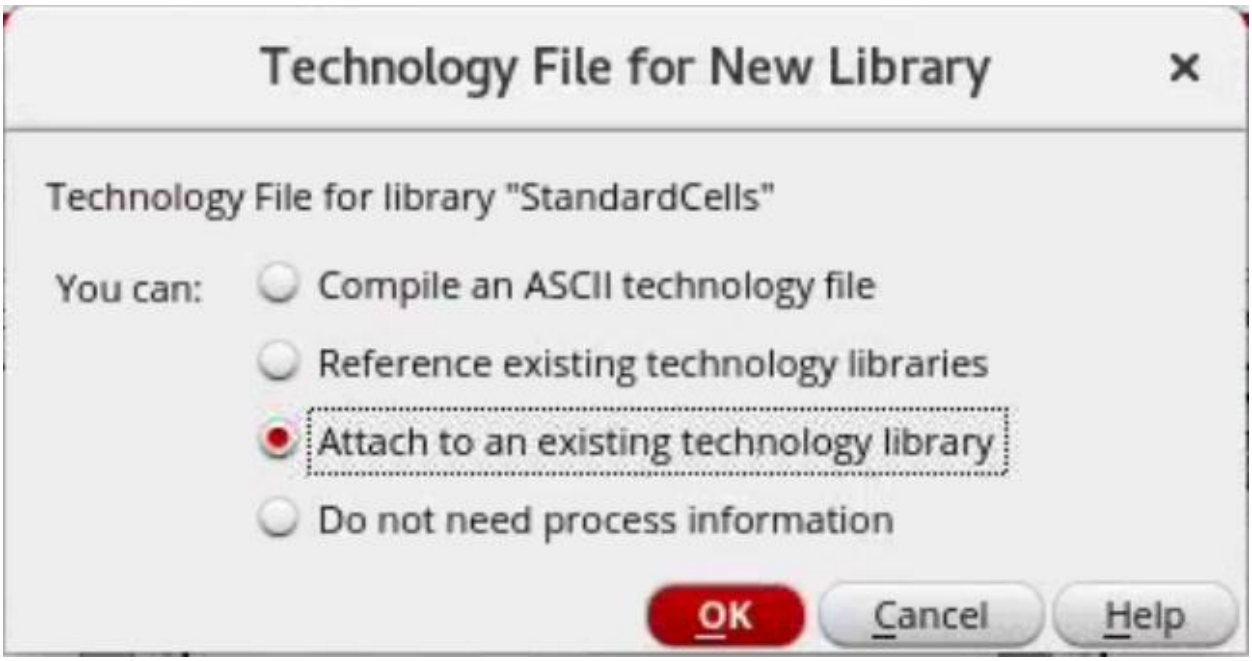
Start Virtuoso:
virtuoso&

Create New Library

File->New->Library



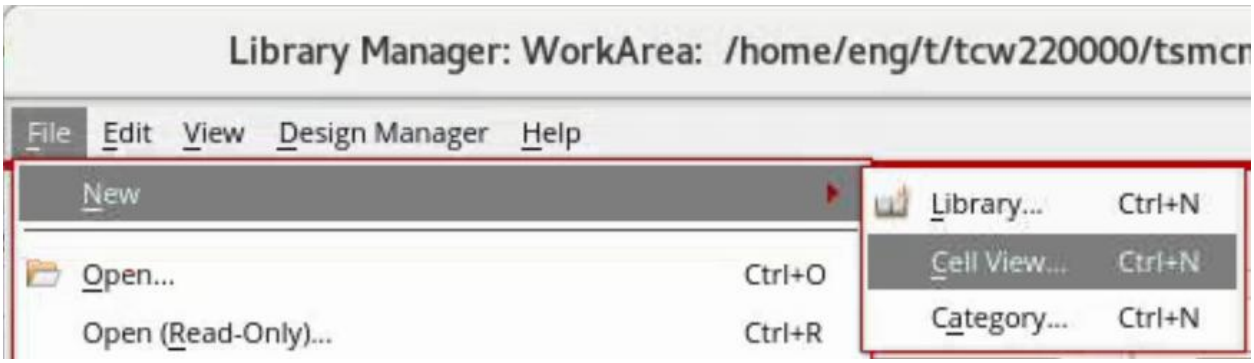
Name your **Library**
Attach to an existing technology library
Select **tsmcN16**
Click OK





Create New Cell View

File->New->Cellview



In **File**:

Library: Select name of the library which you create in previous step

Cell: name of the standard cell you will be designing

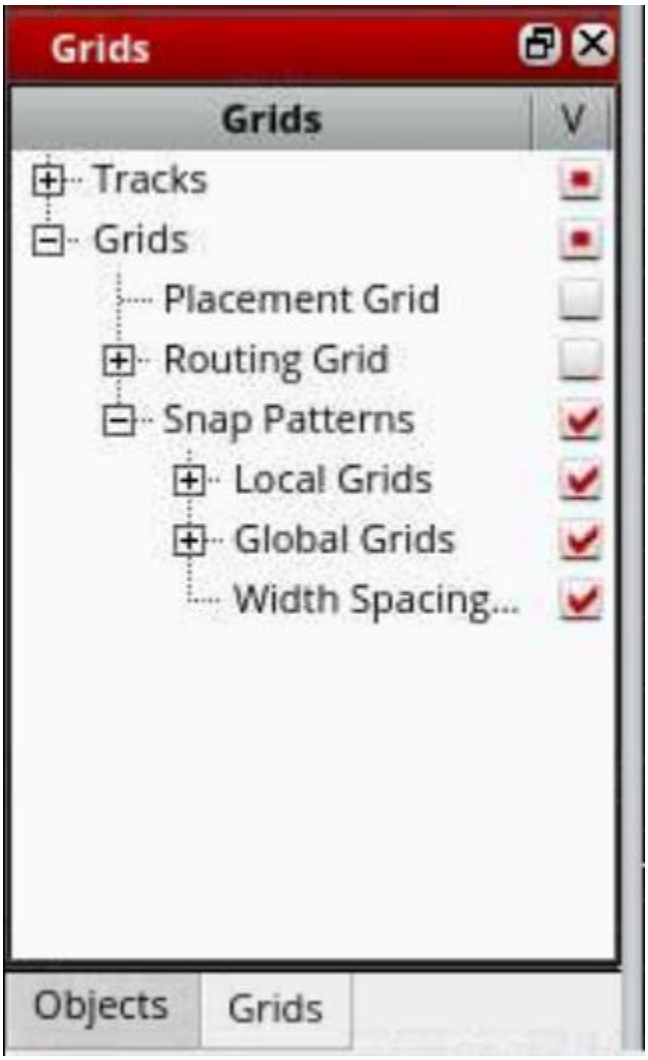
Type: select **layout**

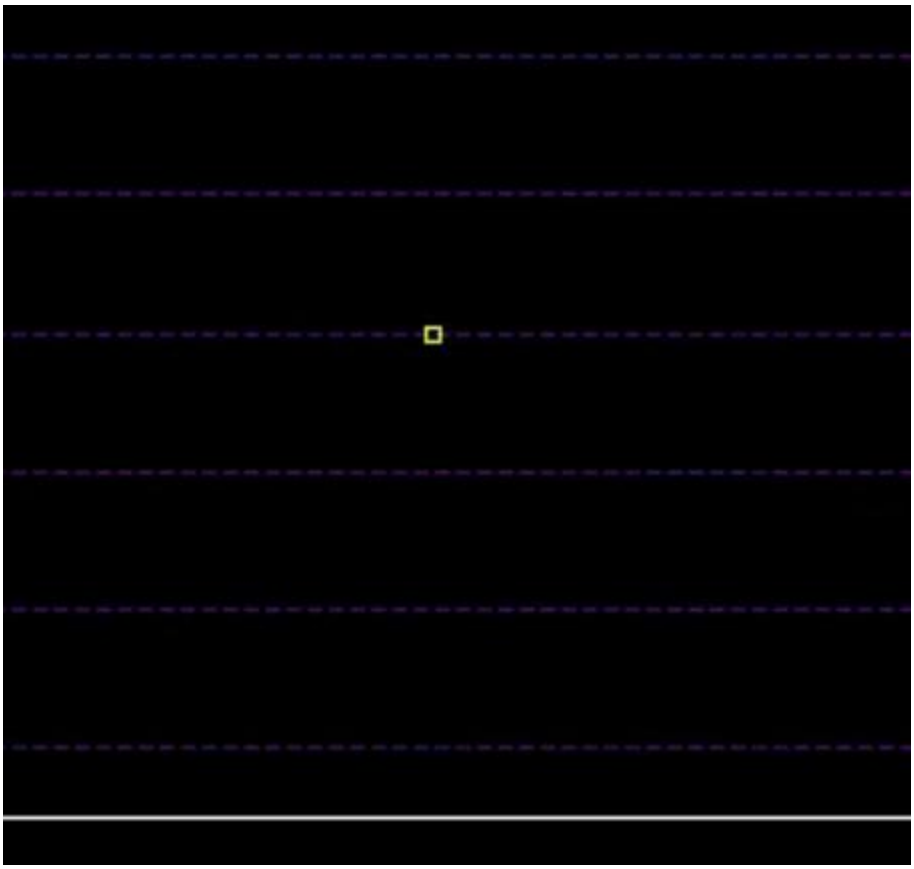


Click **OK**

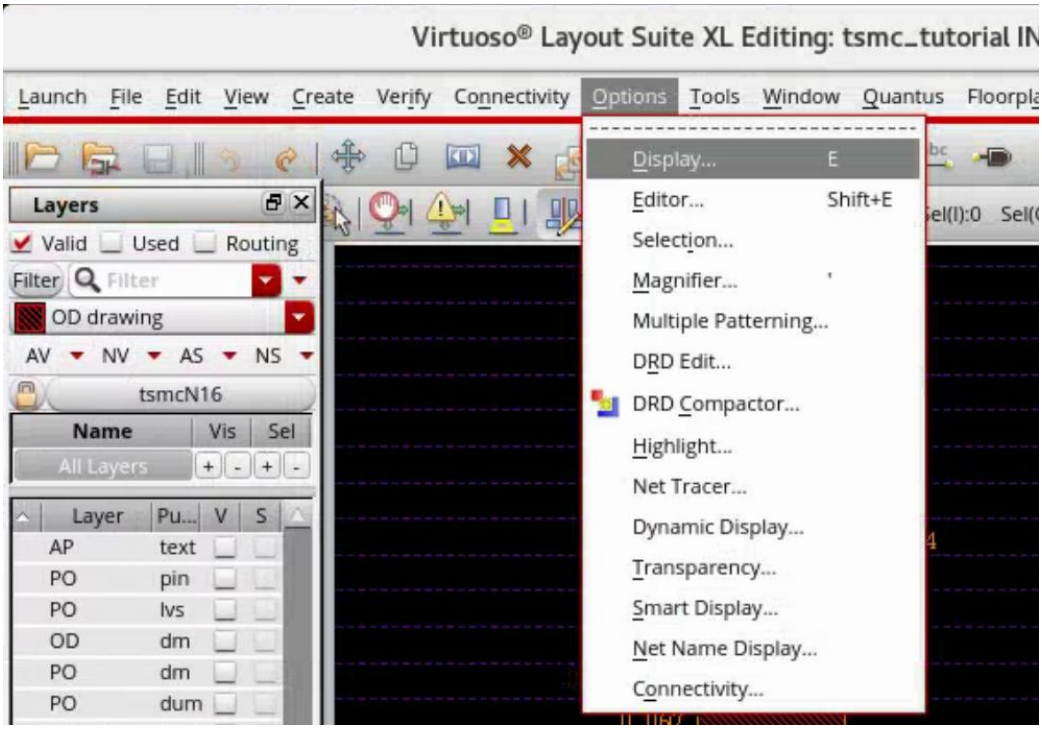
If there is a popup click always.

On the left side pane under **Grids->Snap Patterns->Global Grids**
Make sure GFG is check marked. Zoom in till you see the purple lines. These represent the vertical Fin pitch.

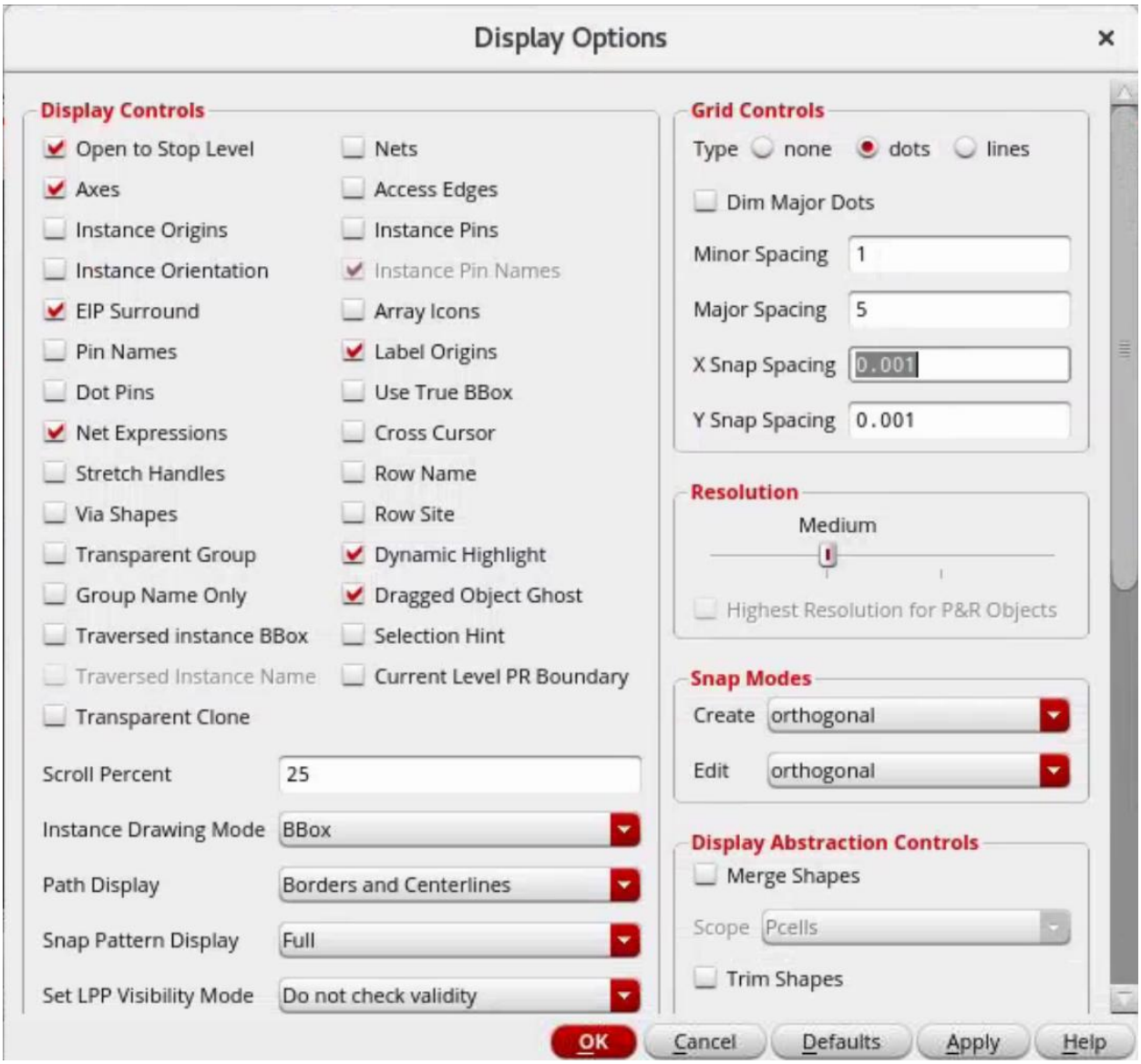




Under **Options->Display->Grid Controls**



X Snap Spacing set to **0.001**
Y Snap Spacing set to **0.001**



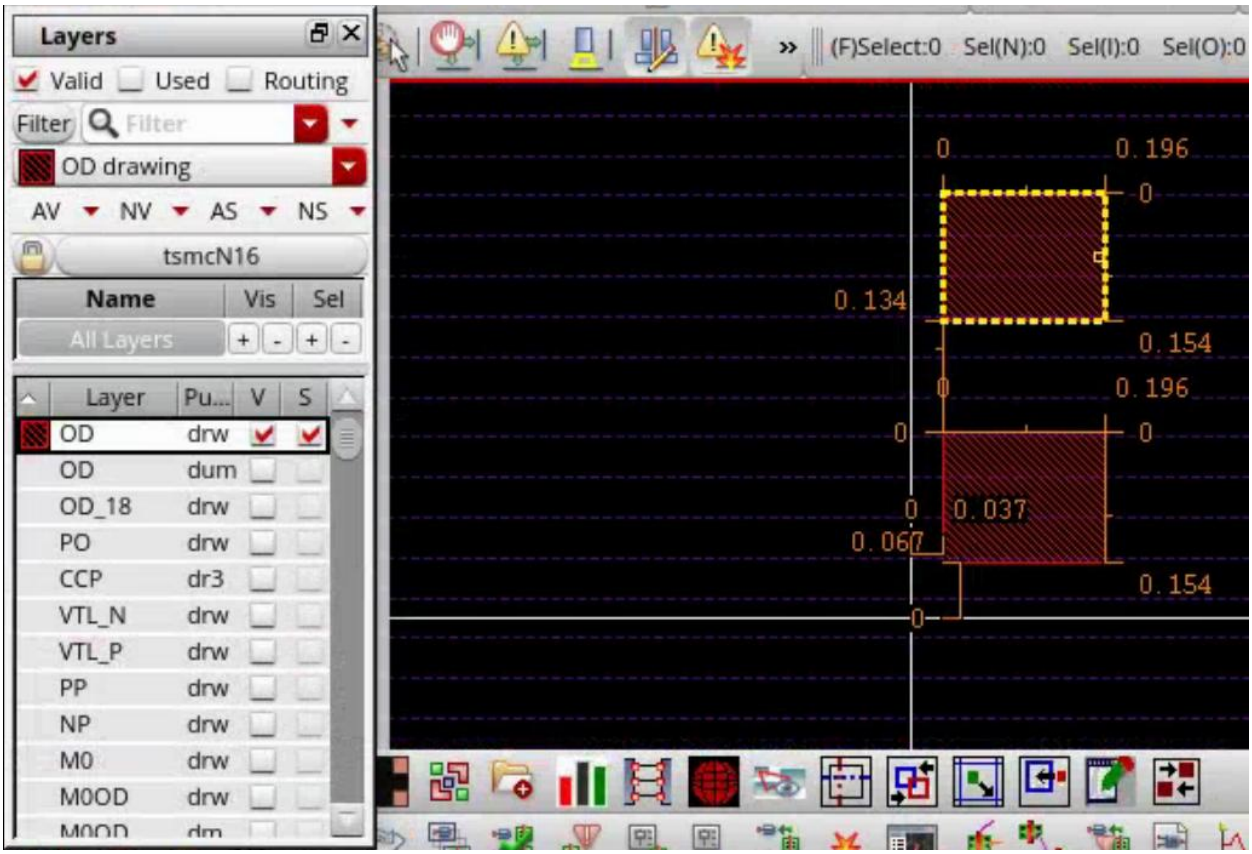
Click **OK**

Inverter Layout Tutorial

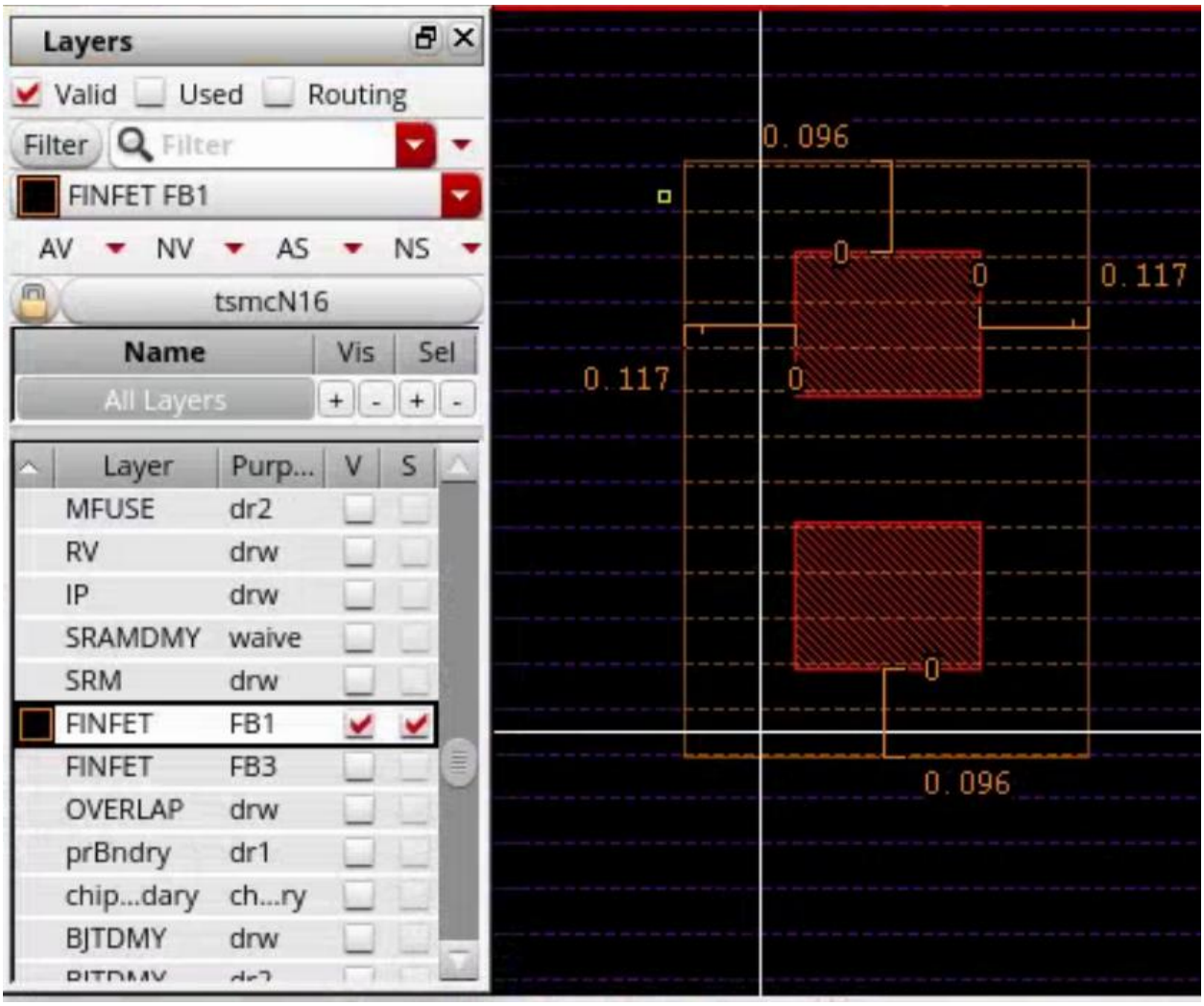
Layout Rules:

- Poly Channel Height = 16nm
- Poly Channel Width = 576nm
- Poly Pitch = 90nm
- Ref Height = 10nm
- Ref Width = OD Width
- Ref Pitch = 48nm
- M0OD Width = 40nm
- Cell Height = 576nm
- Minimum width of M1, M2, M3 =
- V0, V1, V2 = 32nm/32nm

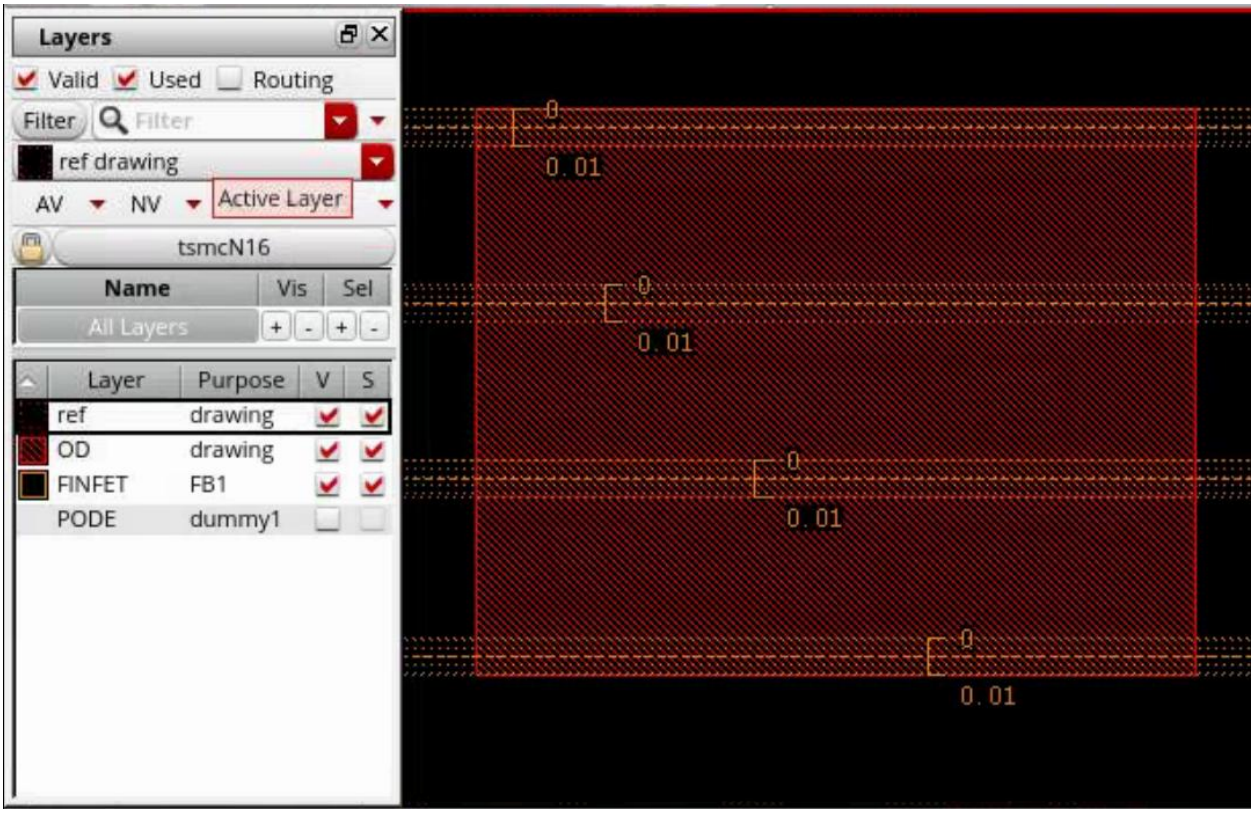
Select **OD drw** layer and press R to draw the layer. Draw two rectangles from the origin of size 196nm and 154nm. Draw the first rectangle 37nm to the right of the origin and 67nm up from the origin. Then draw the second rectangle 134nm above the first.

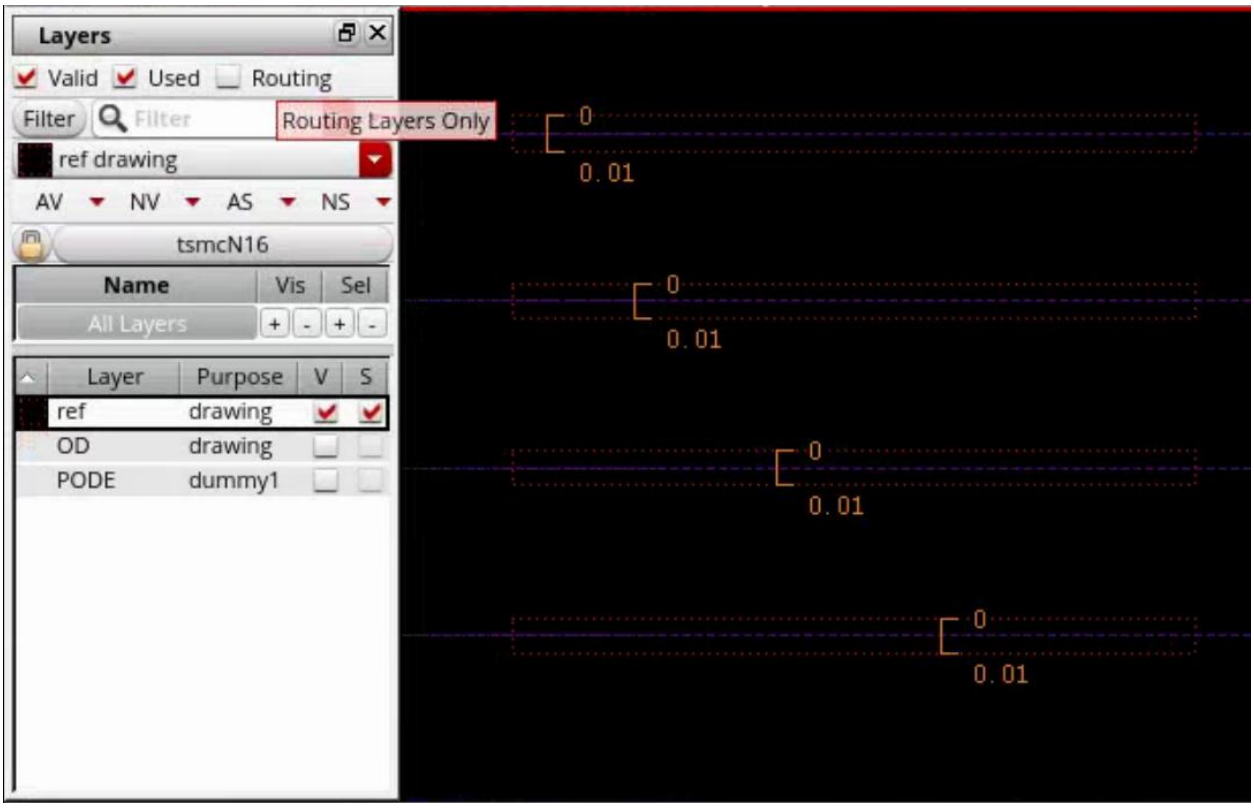


Select **FINFET FB1 layer** and draw the layer symmetrically 117nm left and right of your **OD layer rectangles** and 96nm above and below. The layer snaps to the fin pitch making vertical dimension placement easier.



Select the **ref drawing layer** and create your fin references by drawing 10nm high rectangles on the **OD layer**. These span each fin as defined by the **FINFET FB1 layer**. FB1 follows a height of 250nm+48nm*n where n is the number of fins. It must enclose the OD layer by 96nm+48nm*n where n is number of fins. Both n's are not necessarily the same number. In our case the enclosure area is 4 fins, yet the height is 14 fins total.

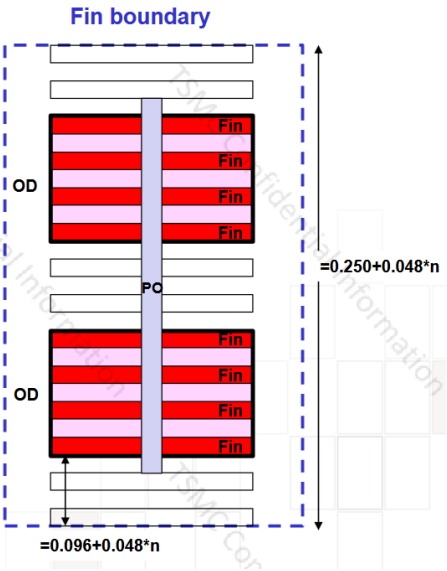




Both the FINFET FB1 and the Ref drawing layers define the fin boundary and fin grids.

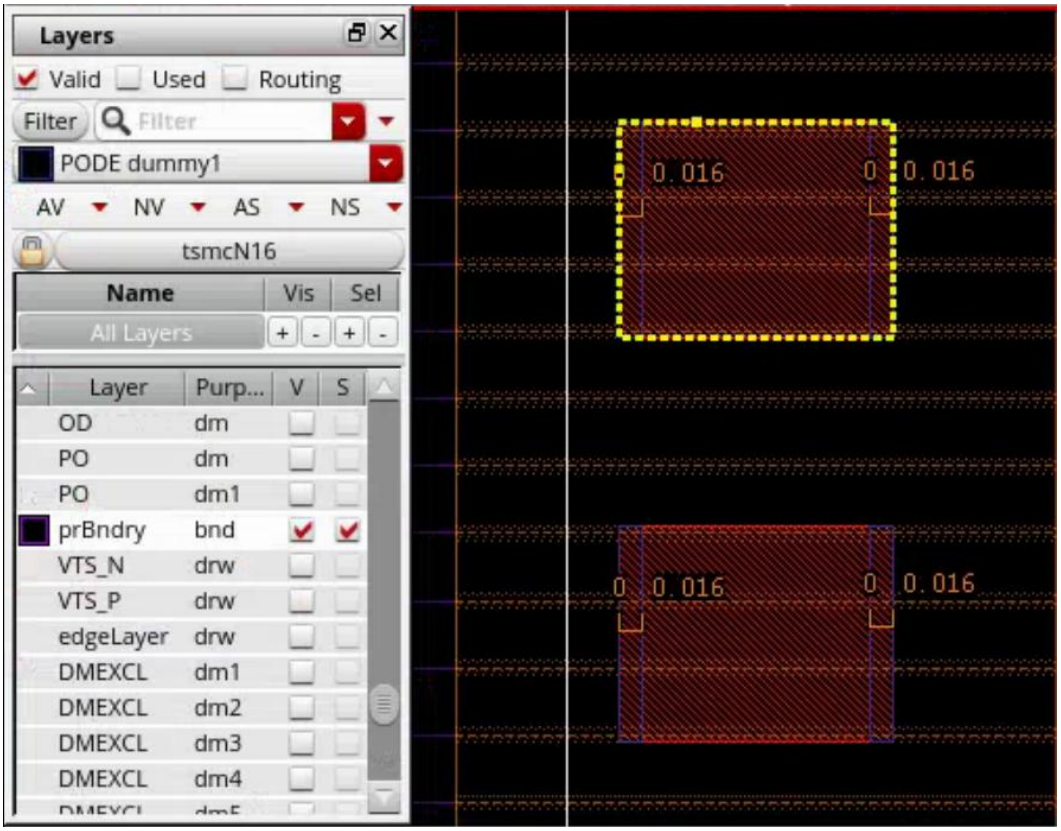
Fin boundary and Fin grid

- **Must follow Fin grid globally**
- **Fin Boundary layer**
  - **FinFET Boundary_1 layer (FB_1 250;0)**
    - On Fin grid
- **Fin_Boundary width (vertical direction)**
  - **Must follow Fin pitch and grid**
  - **0.25 + 0.048*n**
- **Enclosure of OD by Fin_Boundary (vertical direction)**
  - **Must follow Fin pitch and grid**
  - **0.096 + 0.048*n**

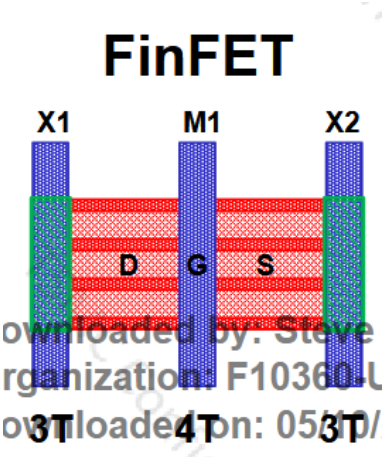


ownloaded by: Steve Martindell
rganization: F10360-University of Texas at Dallas

Select the **PODE dummy1 layer** and draw a rectangle along the edges of the **OD layer**. The rectangle will be 16nm wide and will always match the height of the **OD layer** except for specific use cases.

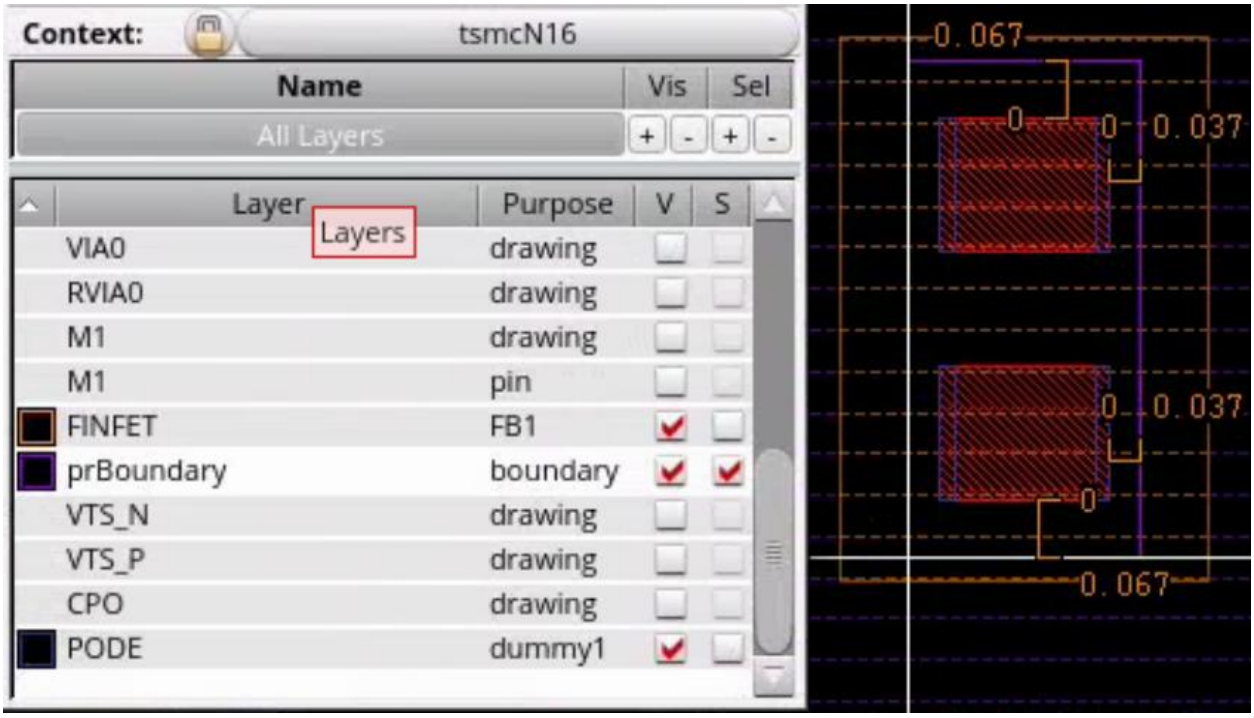


This layer is defined as the **Poly on Oxide Diffusion Edge layer**, or **PODE layer**. For our case, it defines a 3T (3 terminal) device, typically, these are devices at the edge of the layout.

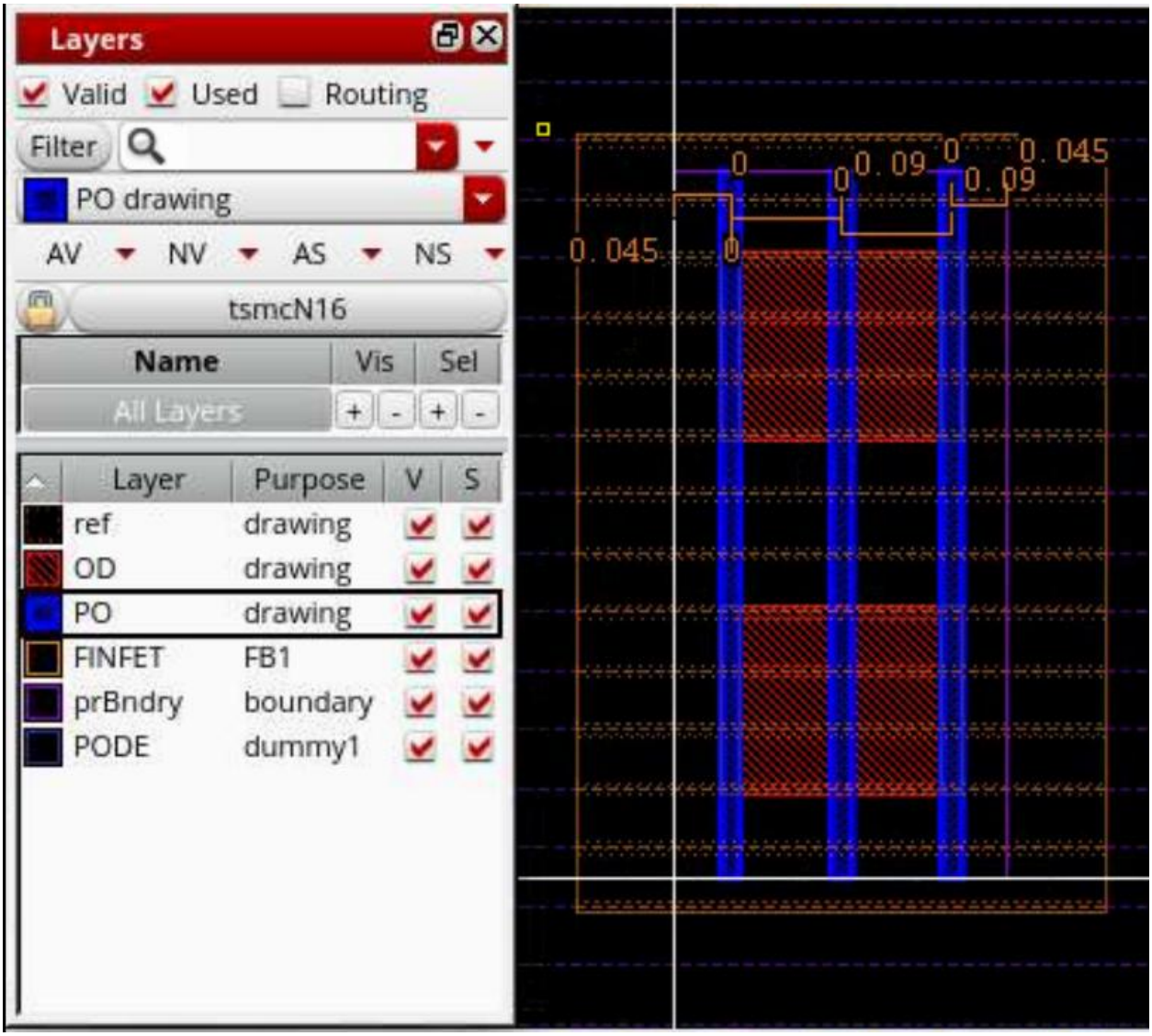


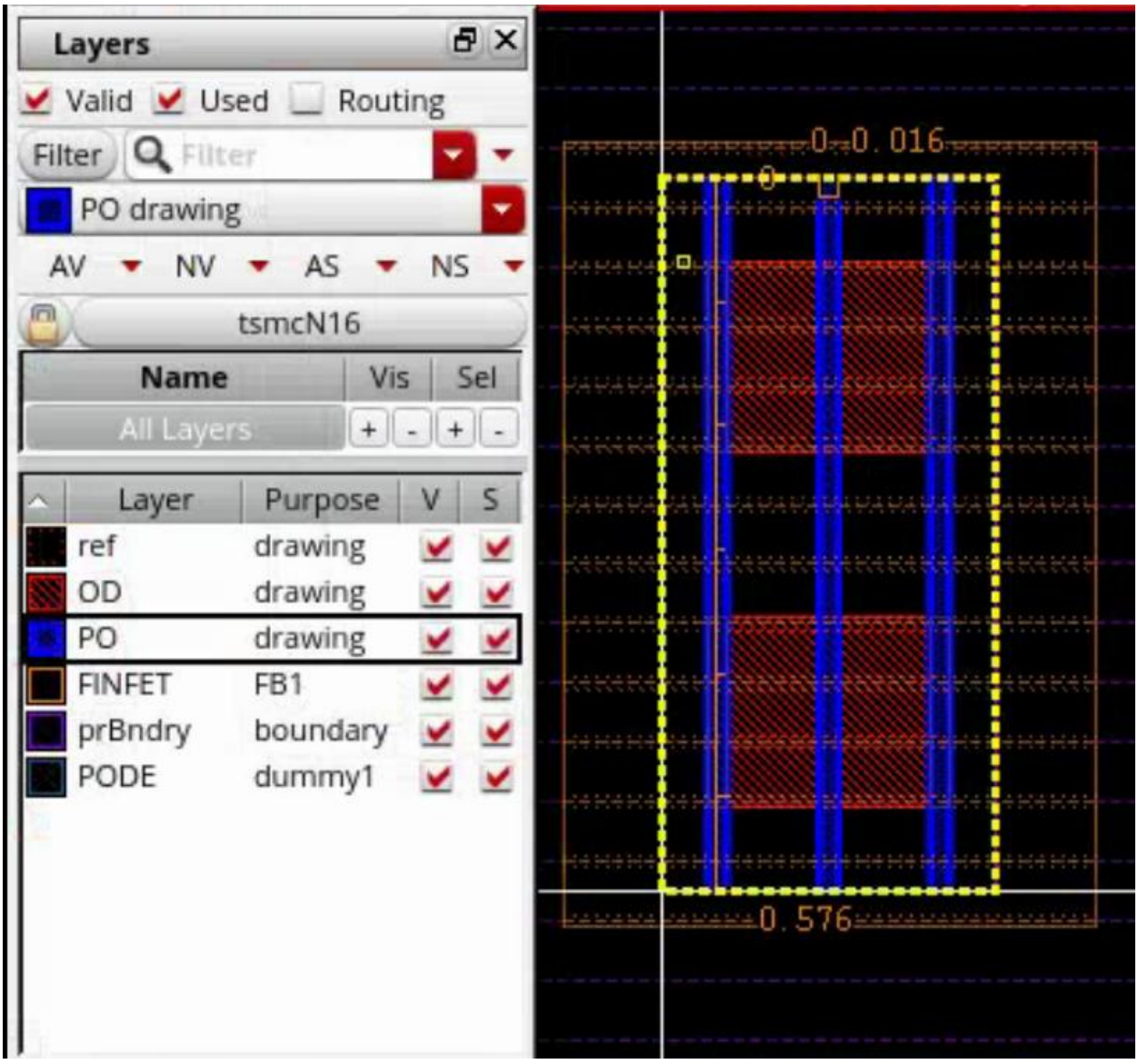
ownloaded by: Steve
rganization: F10360-L
ownloaded on: 05/2/

Now draw the **prBndry boundary** layer. This layer will always be 576nm in height or 67nm from the bottom and top of the **OD drawing layer**. The width of the layer is always 37nm from the left and right of the **OD drawing layer**. This layer dictates the place and route area for auto place and routers such as **Innovus**. In a sense this is your true height and width of the cell.



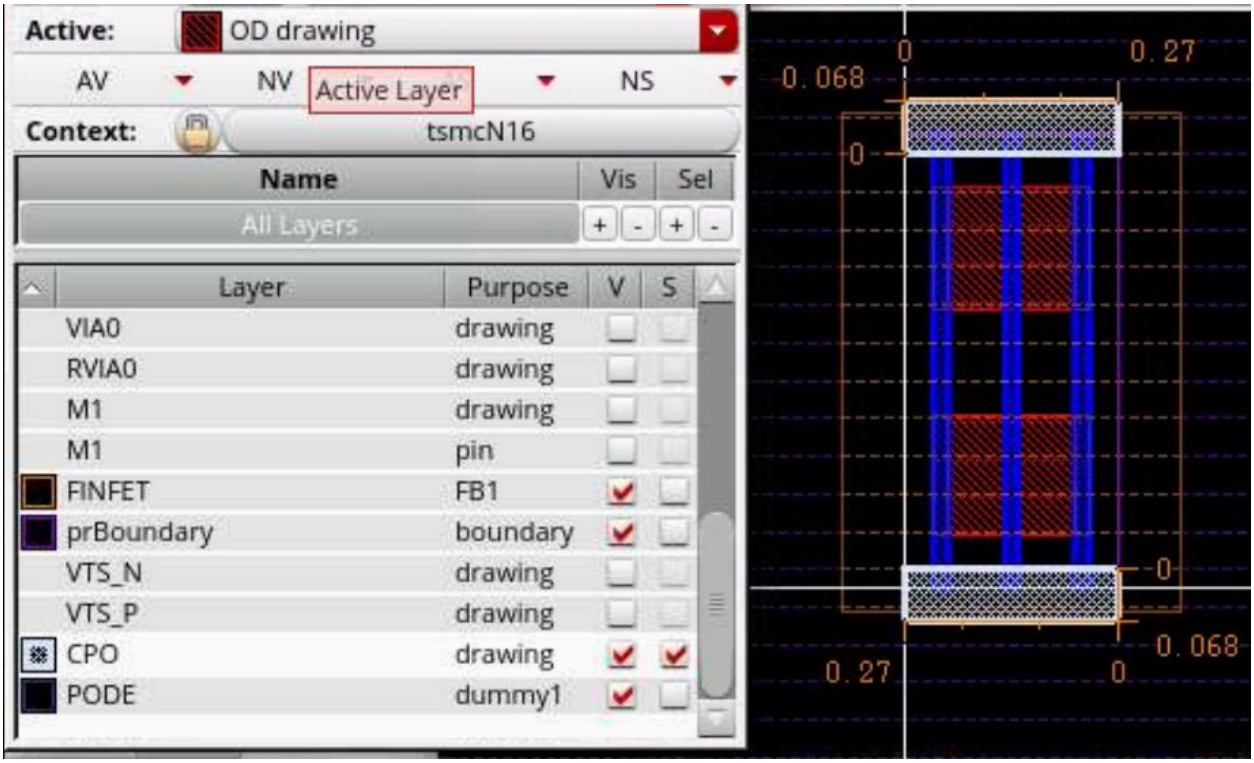
Now draw the **PO drawing layer**. This is your Poly/Gate which will create your 3T or 4T devices. The height is 576nm or the same as the **prbndry layer**. As the name of the technology suggests, the width is 16nm. Poly has a pitch of 90nm from middle to middle of each poly. Notice that the middle of the poly to the edge of the prboundary is 45nm. This is intentional as auto place and routers will place each cell side by side to create a continuous poly pitch of 90nm.

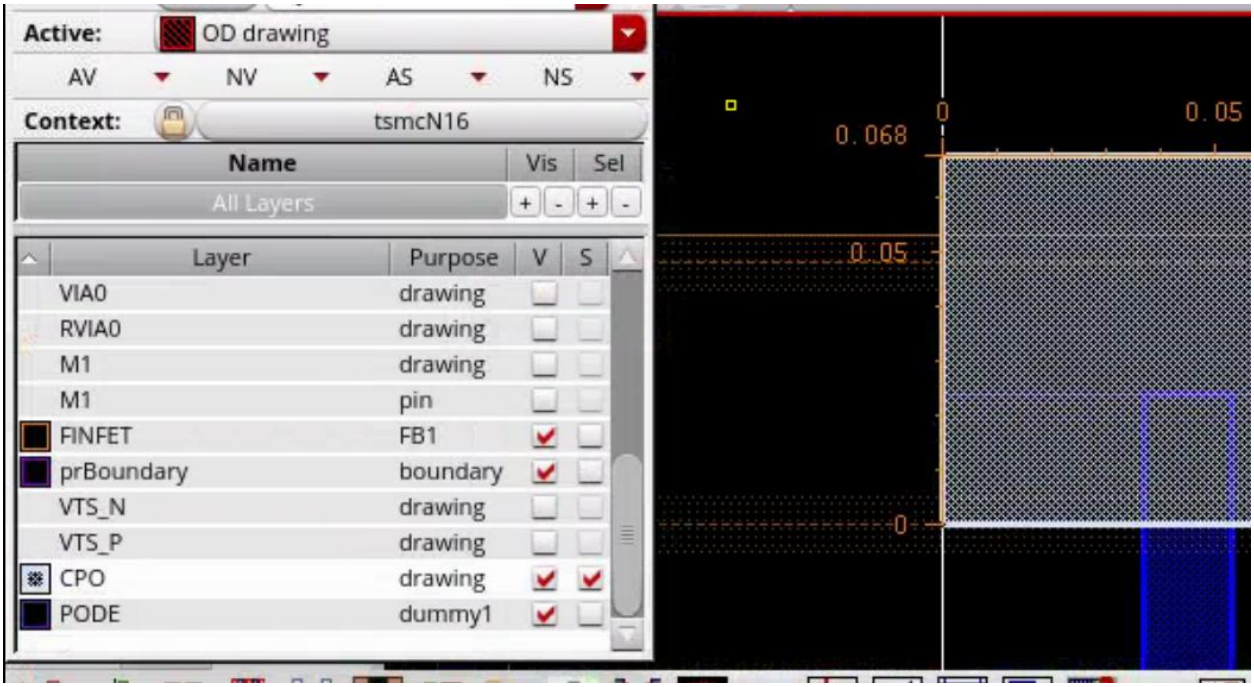




- Poly Width = 16nm
- Poly Height = 576nm
- Poly Pitch = 90nm

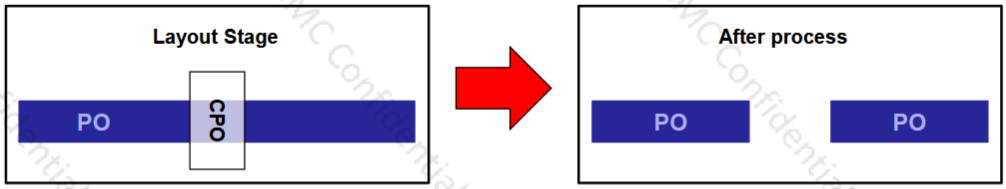
Lastly for Poly we will add **CPO drawing layer**. This layer is the **Cut-POly layer**, and this layer splits or "cuts" the poly without having to space them accordingly when the cells are stacked vertically later during auto place and route. The layer spans the width of your cell (prboundary layer) and is 68nm tall place directly from middle to middle of fins from the **FINFET FB1 layer**.



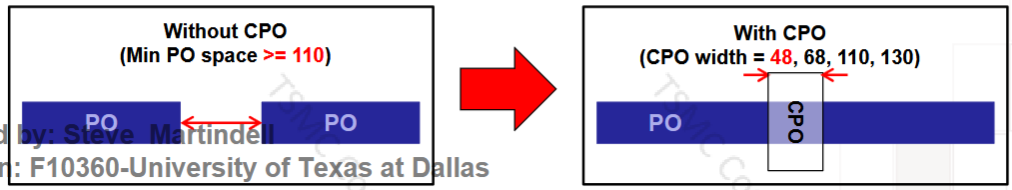


Cut-Poly introduction (CPO)

- **CPO layer can split PO**

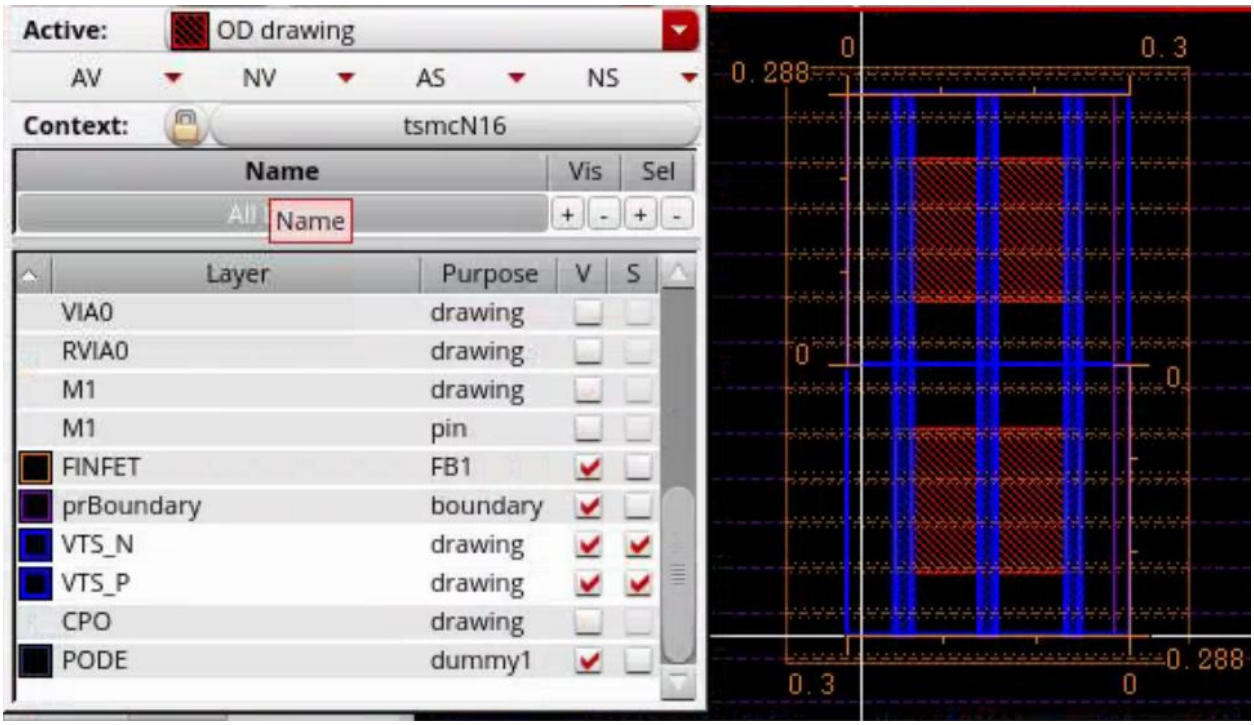


- **PO E2E space can be less w/ CPO layer (110nm vs 48nm)**

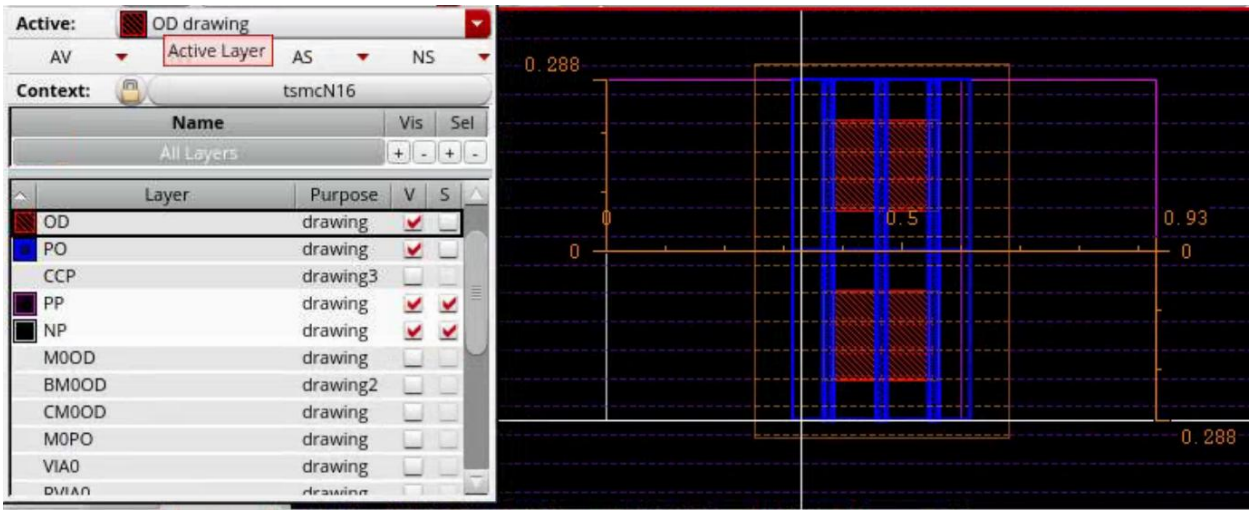


wnloaded by: Steve Martindell
ganization: F10360-University of Texas at Dallas
wnloaded on: 05/10/2025

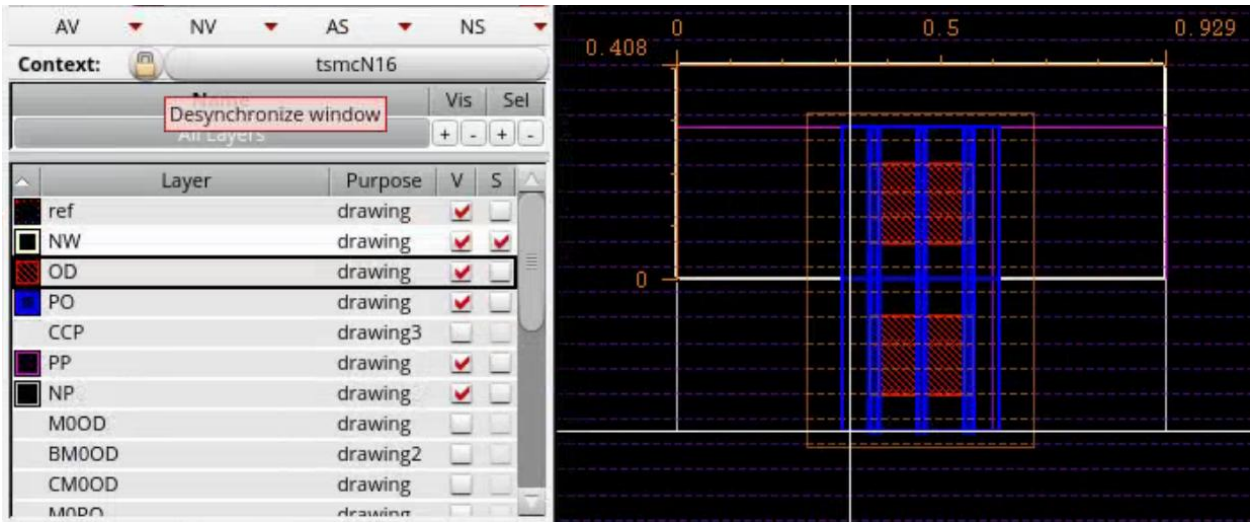
Draw the VTS layers, **VTS_N** and **VTS_P drawing layers**. The VTS layers are used to determine what type of voltage is used for the cell. In this case, it can be standard voltage threshold (svt) or low voltage threshold (lvt). For our purposes we will create svt cells. **VTS_N** will cover the nmos side of the cell, and the **VTS_P** will cover the pmos side of the cell within the prboundary. This layer should be 300nm wide and 288nm tall for each one.



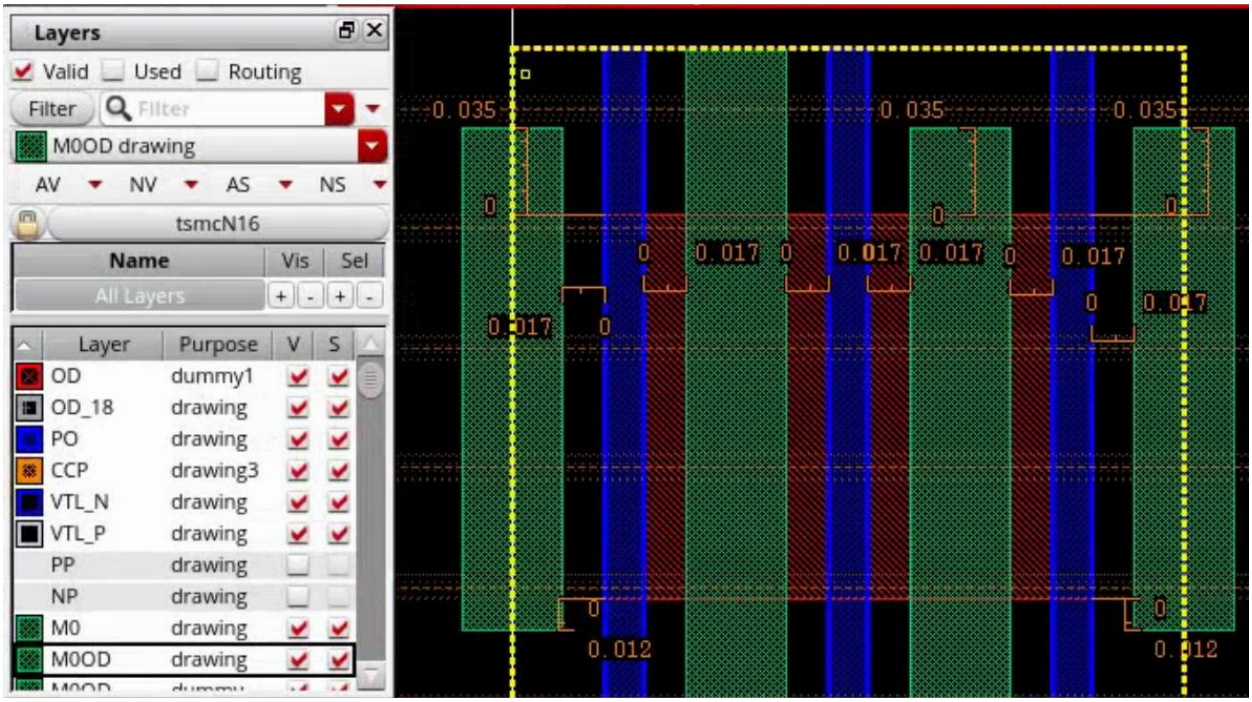
Now the **NP drawing layer** and the **PP drawing layer**. These layers need to be at least 0.24 in area. Extend them horizontally to fit the area requirement. These are the implant layers for N doping and P doping.

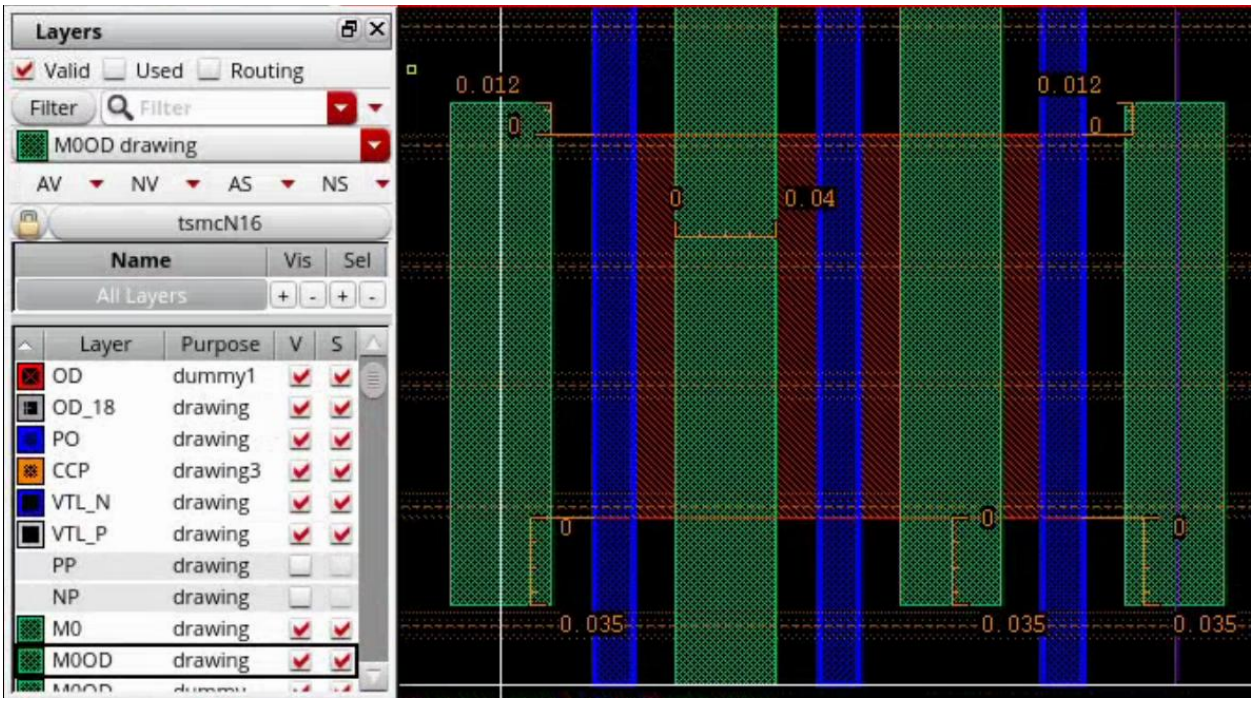


Lastly for implant layers, the **NW drawing layer** or N-Well layer is for your pmos N-Well. The N-Well layer needs to be large enough to fit the area requirement for DRC. The area needs to be at least 0.324 or more.

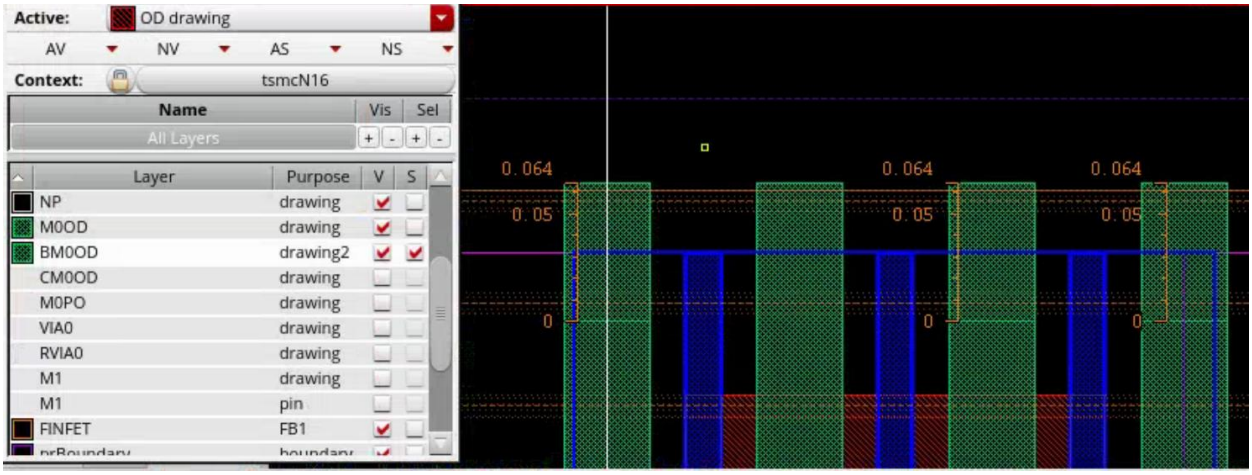


Next, draw the **Metal 0 Oxide Diffusion layer** or **M0OD drawing layer**. Notice this layer locks on specific widths while drawing. Be sure to use the 40nm width. Space each drawing symmetrically between the poly. Where there is **Oxide Diffusion** there will be a connected **M0OD layer**. Whether the **M0OD layer** is cut by the **CM0OD layer** depends on if you are trying to connect the source and drain of a pmos+nmos. However, watch out for a DRC rule concerning **M0PO**. Do not worry about that for now.

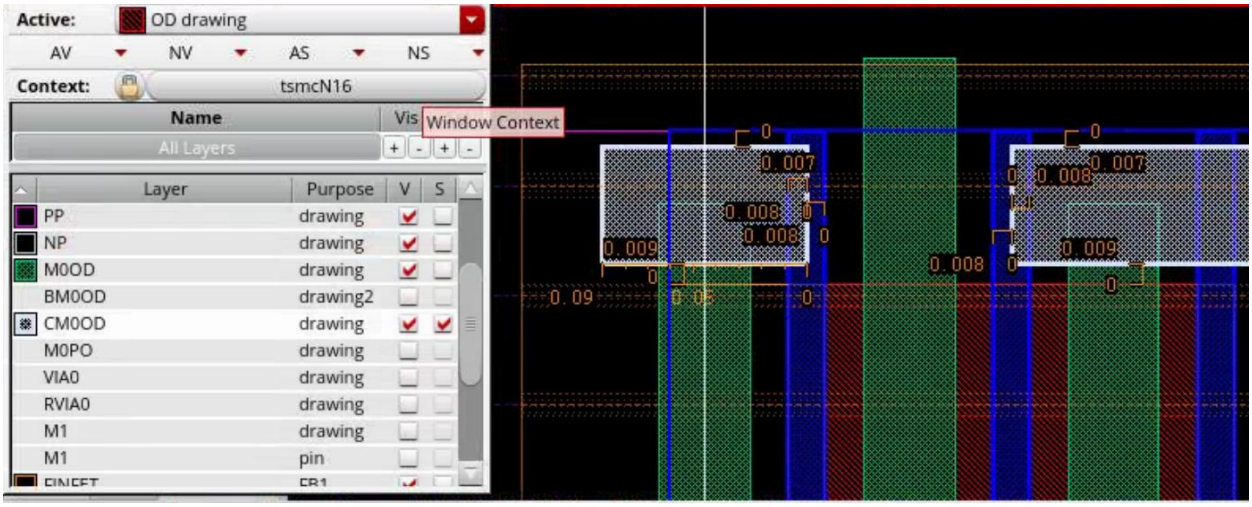


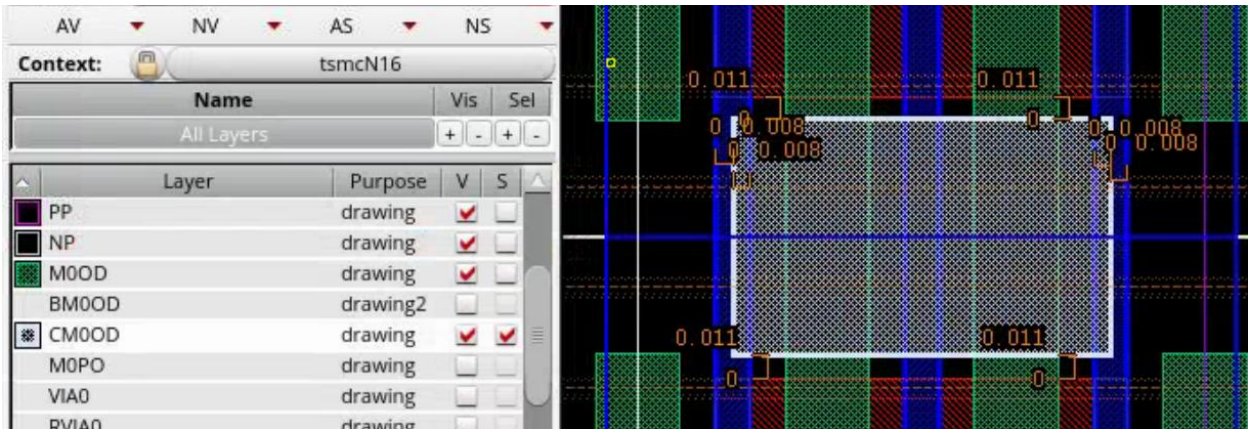


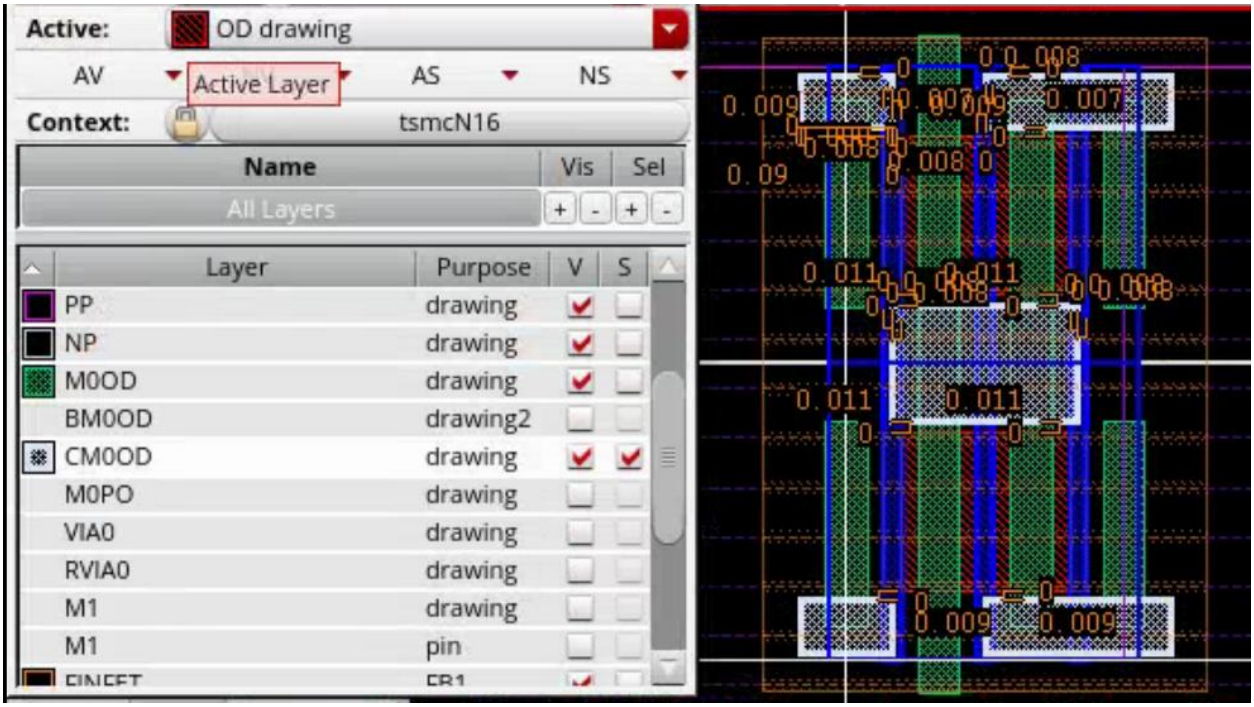
Next, to add to the **BM0OD drawing2 layer** we will be adding what is called an abutment layer for M0OD. These are added to both ends of the **M0OD layer** and are for the tape out process to make it easier to combine M0OD layers.



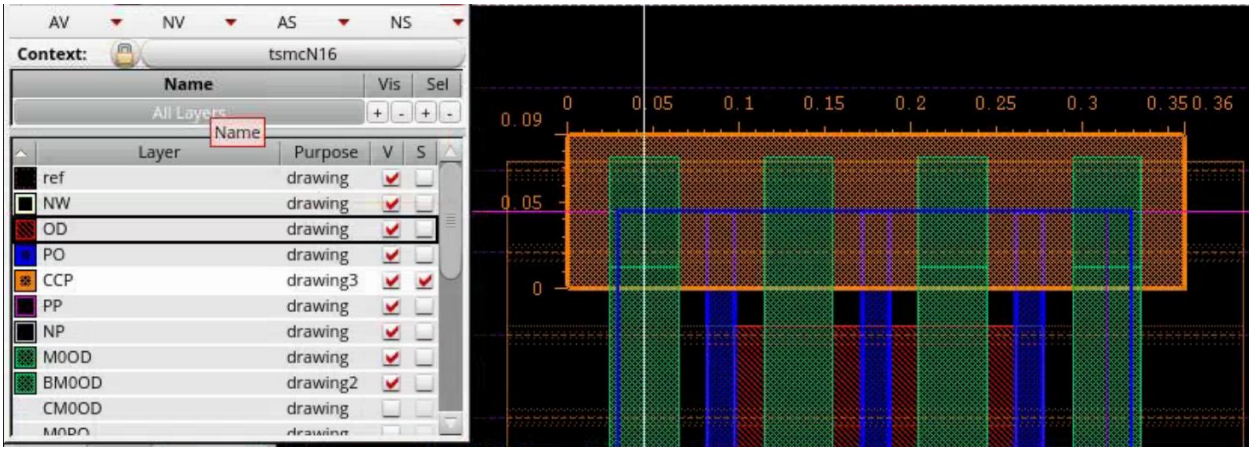
Now draw the **Cut Metal 0 Oxide Diffusion Layer** or **CM0OD drawing layer**. This layer is used to split **M0OD layers** at specific widths less than 96nm. For our purposes the layer will have a vertical width of 112nm. The layer shares a color with **CPO layer** **DO NOT** confuse the two. Leave a gap where the **M0OD** will connect to the VDD/GND of the cell. Be sure to space the edges as if it is connecting to another Poly outside the cell. Leave it hidden afterwards for better visibility.

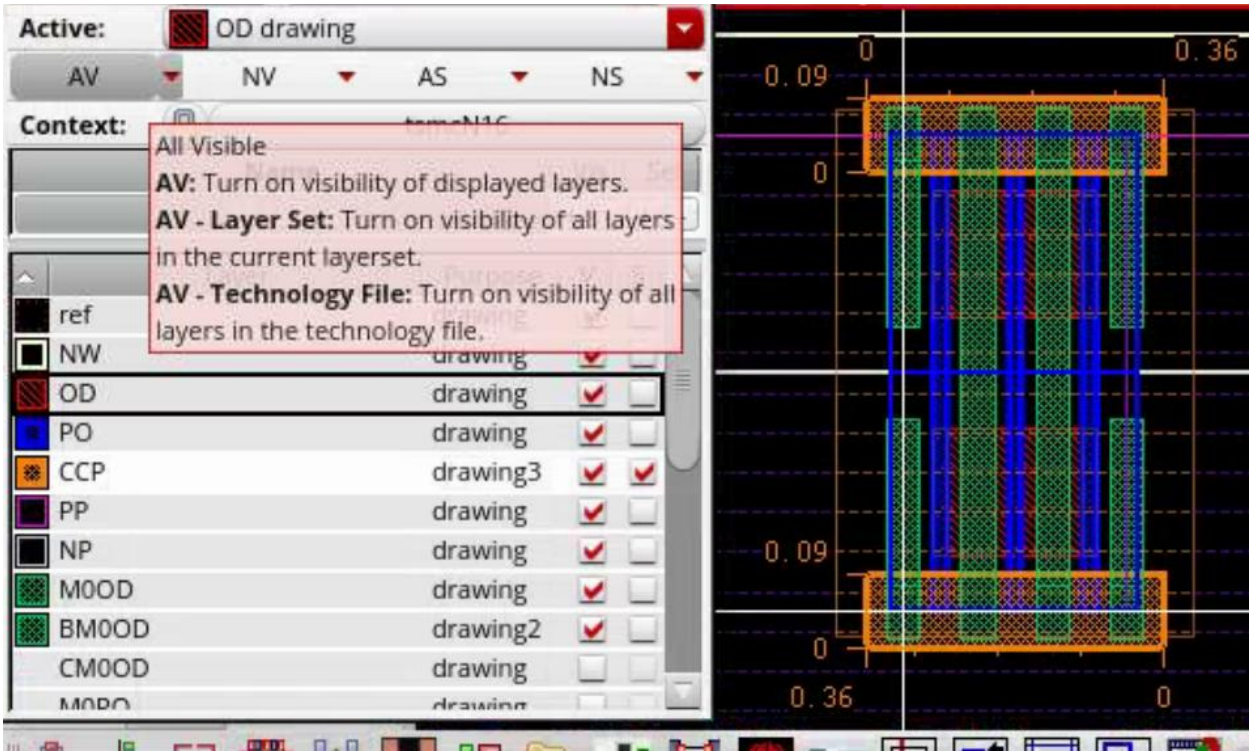



AV
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Name
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BM0OD
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CM0OD
drawing
M0PO
drawing
VIA0
drawing
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Active:
OD drawing
AV
Active Layer
AS
NS
Context:
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Layer
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M0OD
drawing
BM0OD
drawing2
CM0OD
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M0PO
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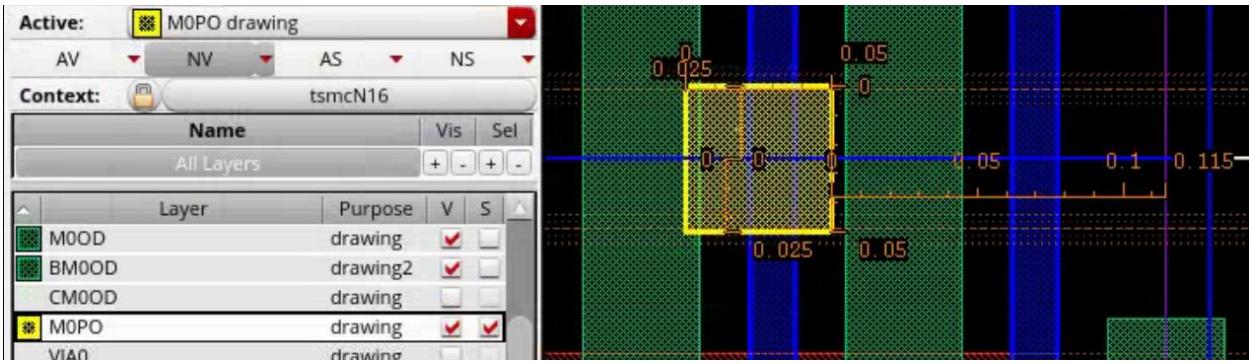
Next, we add the **CCP drawing3 layer**. This layer is for taping out to define density and filling. The layer is 90nm in height and this layer is placed directly on top of your M1 VDD and M1 GND layer. It matches the lengths of the respective M1 layer and must be extended past the boundary layer on both sides.



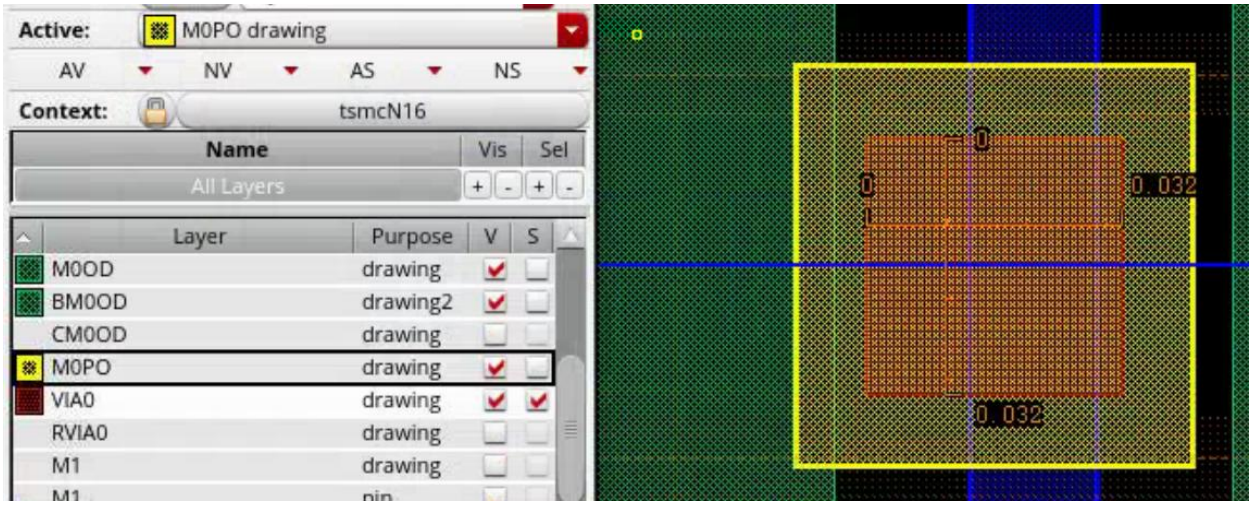


To review we now have **CCP**, **CM0OD**, and **CPO layers** hidden for the continuation of the guide.

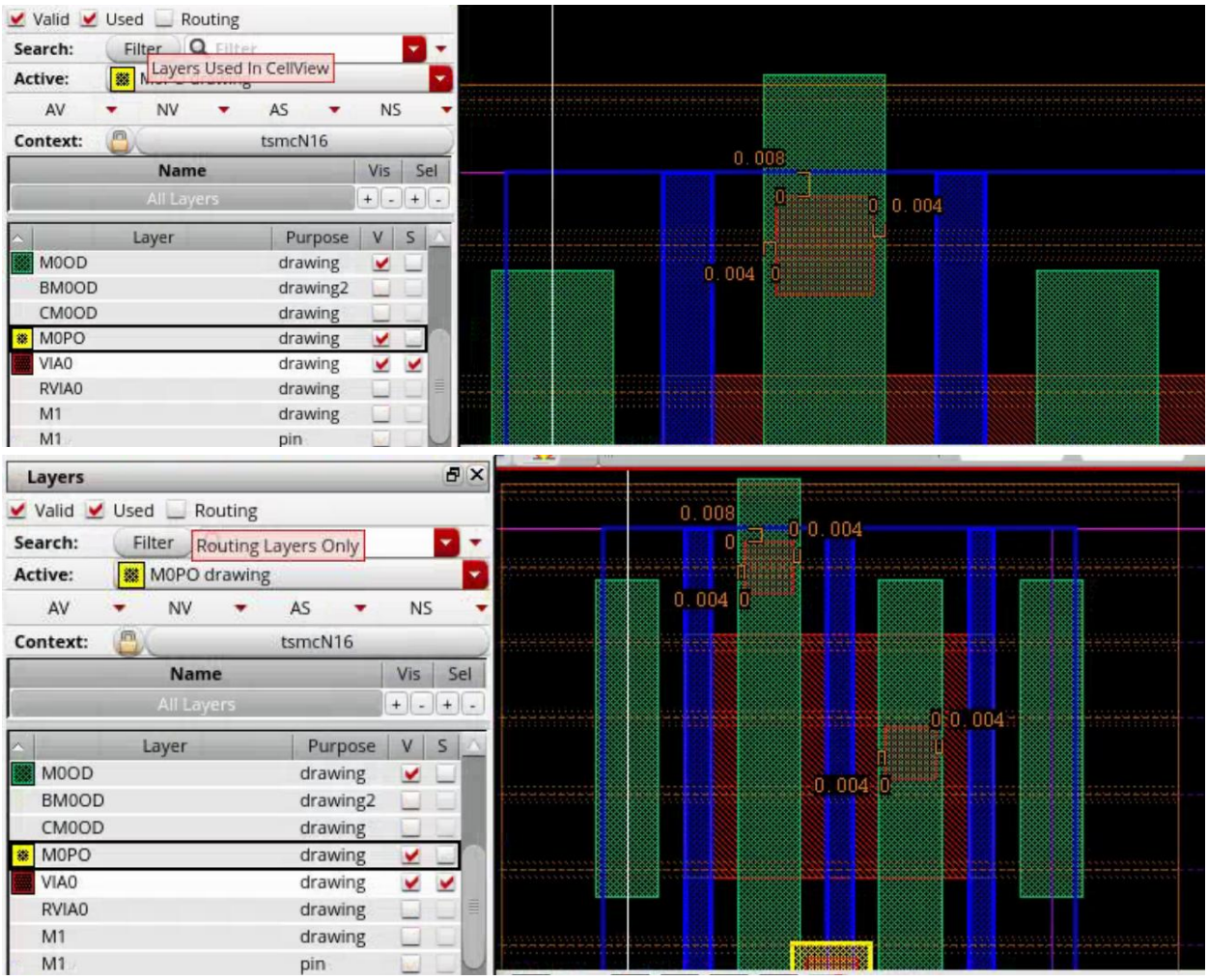
Next, we add the **M0PO drawing layer** or Metal 0 Poly connection layer. This layer is an interconnection layer between **M0OD**, **PO**, and **VIA0 layer**. The Height and Width are both 50nm.

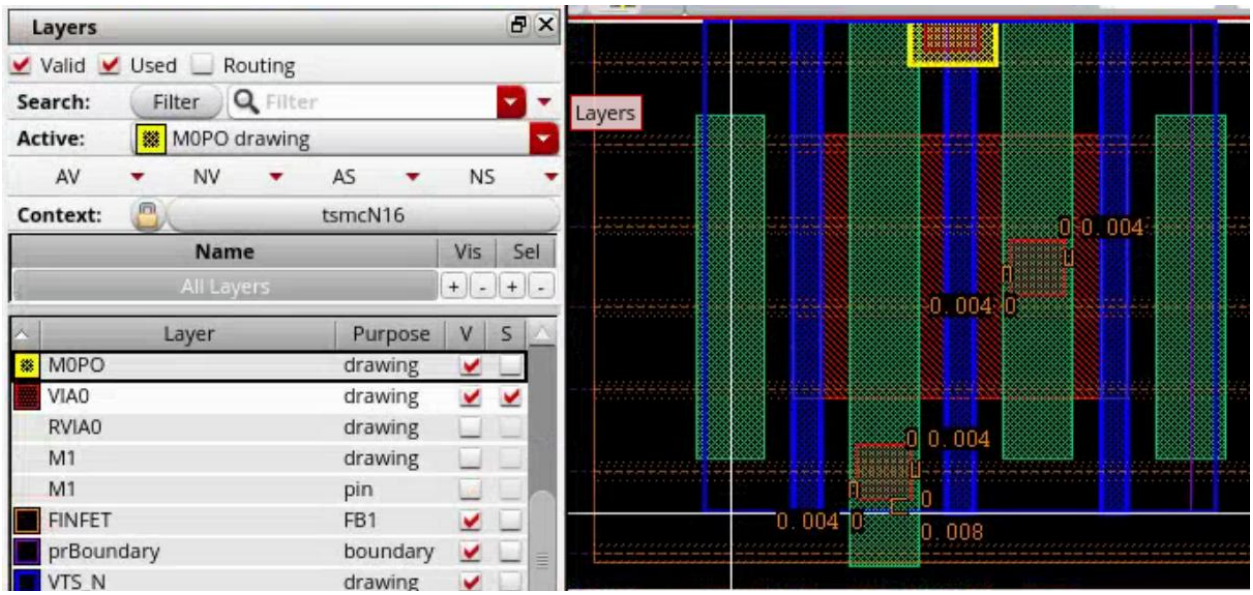


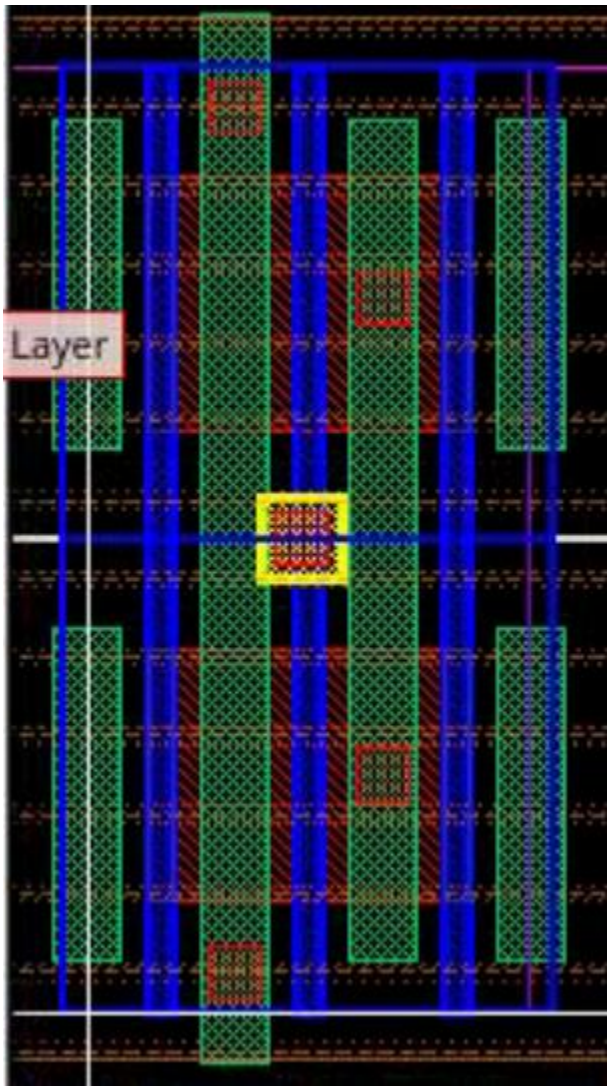
Now place a **VIA0 layer** in the direct middle of the **M0PO layer.** All VIA0s are 32nm by 32nm.



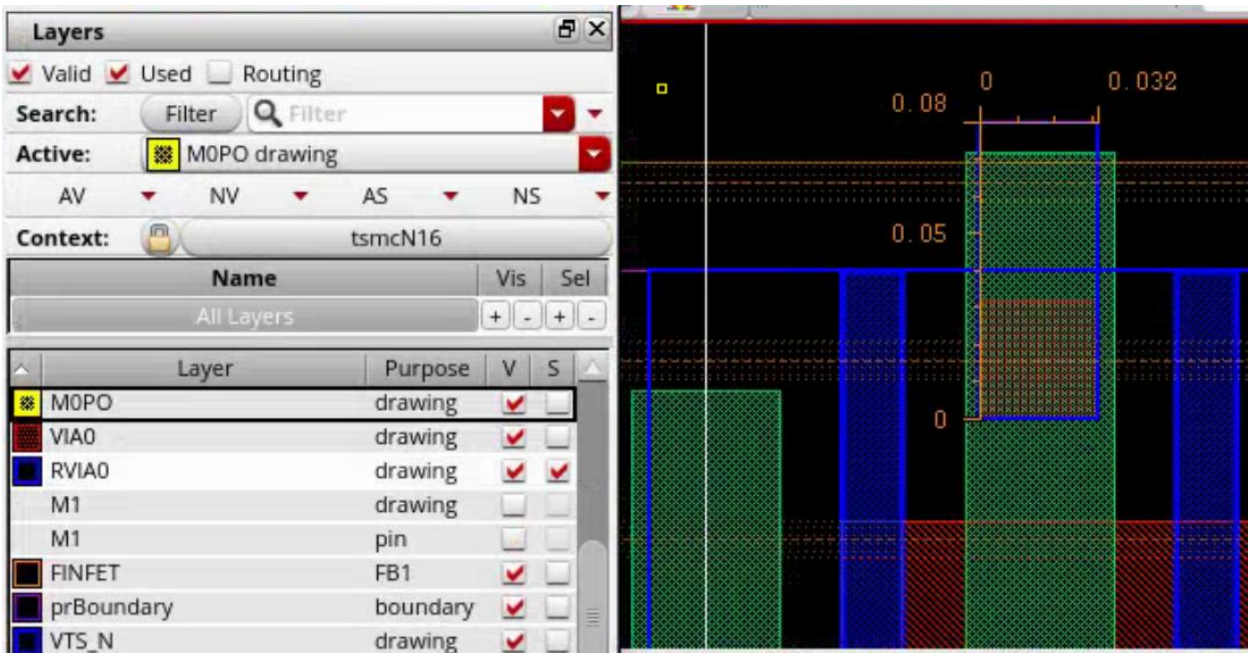
Place the rest of your VIA0s. Since this is an INV they will be mirrored. Two are placed on the PMOS side and two on the NMOS side.

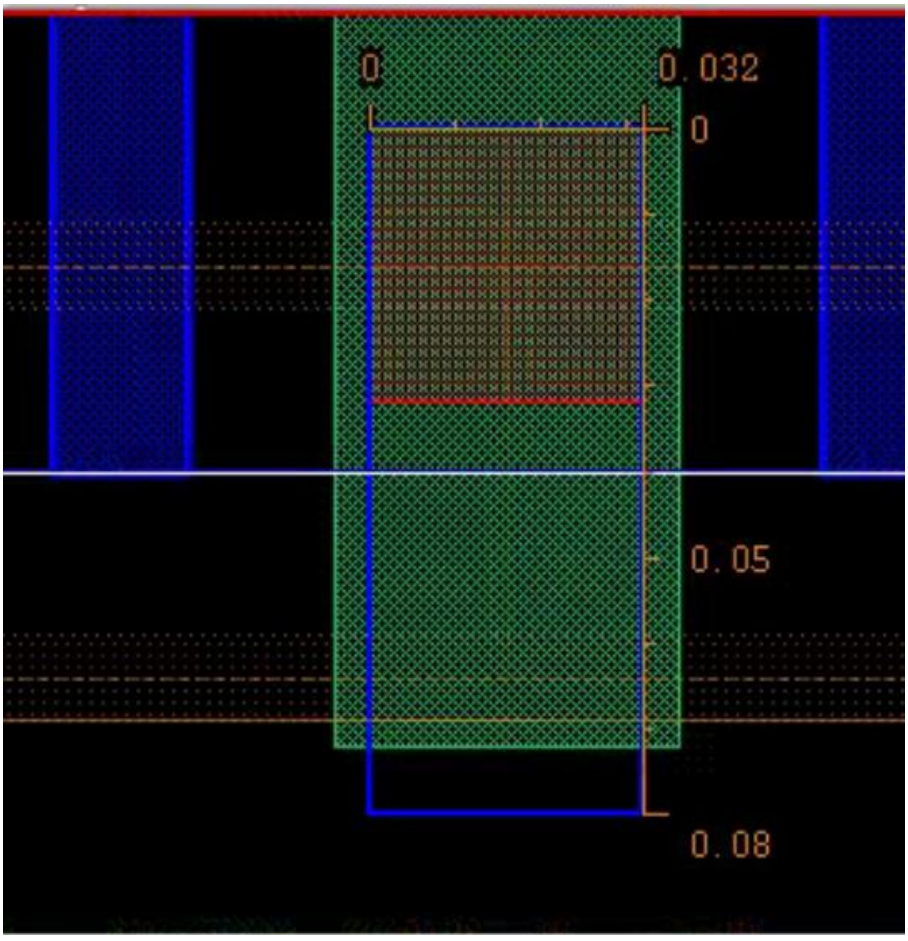



Layers
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M0PO drawing
AV
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NS
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tsmcN16
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Layer
Purpose
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S
M0PO
drawing
VIA0
drawing
RVIA0
drawing
M1
drawing
M1
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0.004
0.004
0.004
0.008

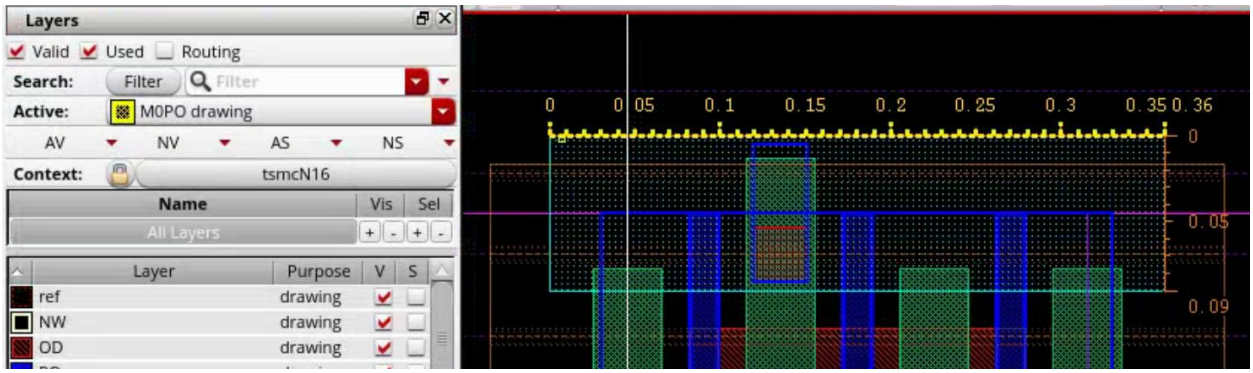

Layer

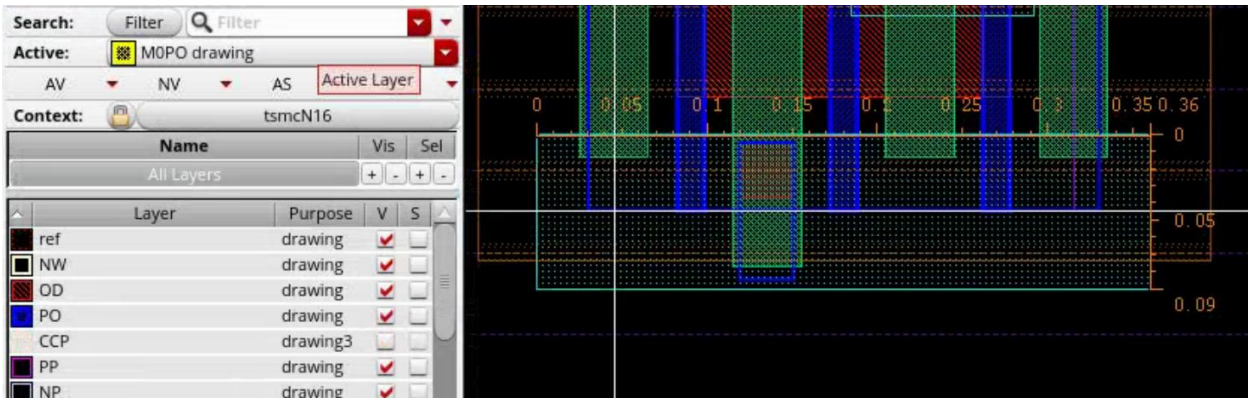
Add the **RVIA0 drawing layer** at the ends of the cell. This allows the tape out process to combine vias into one full long via rather two separate **VIA0s**.



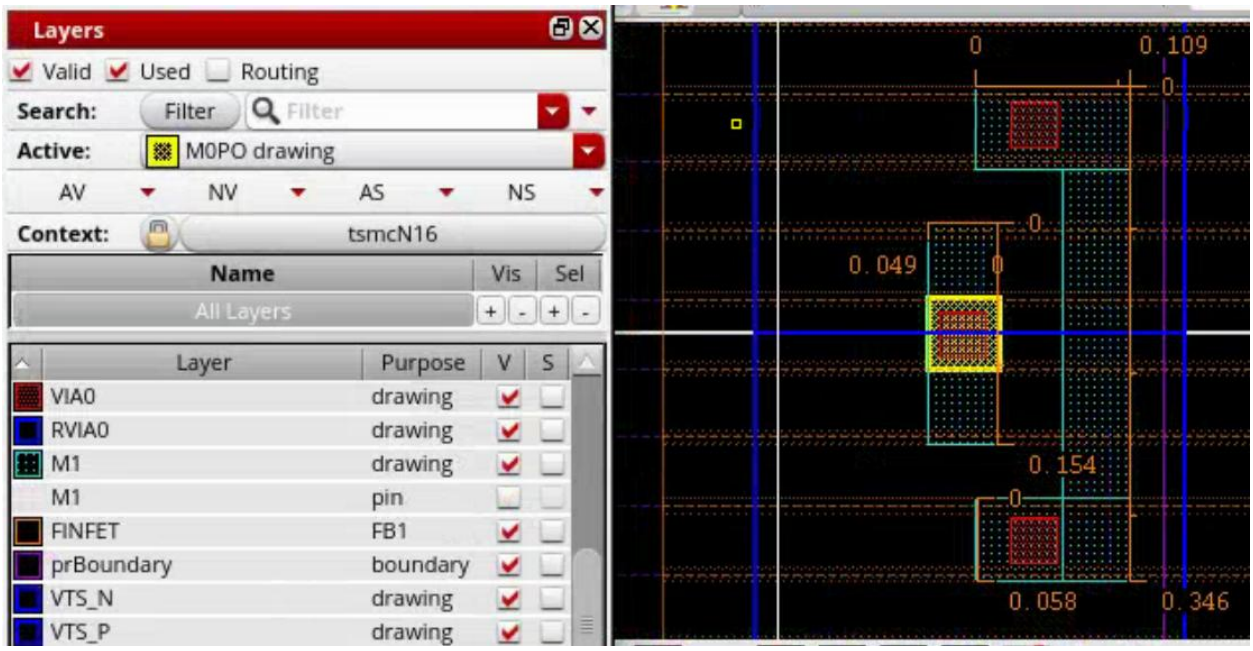


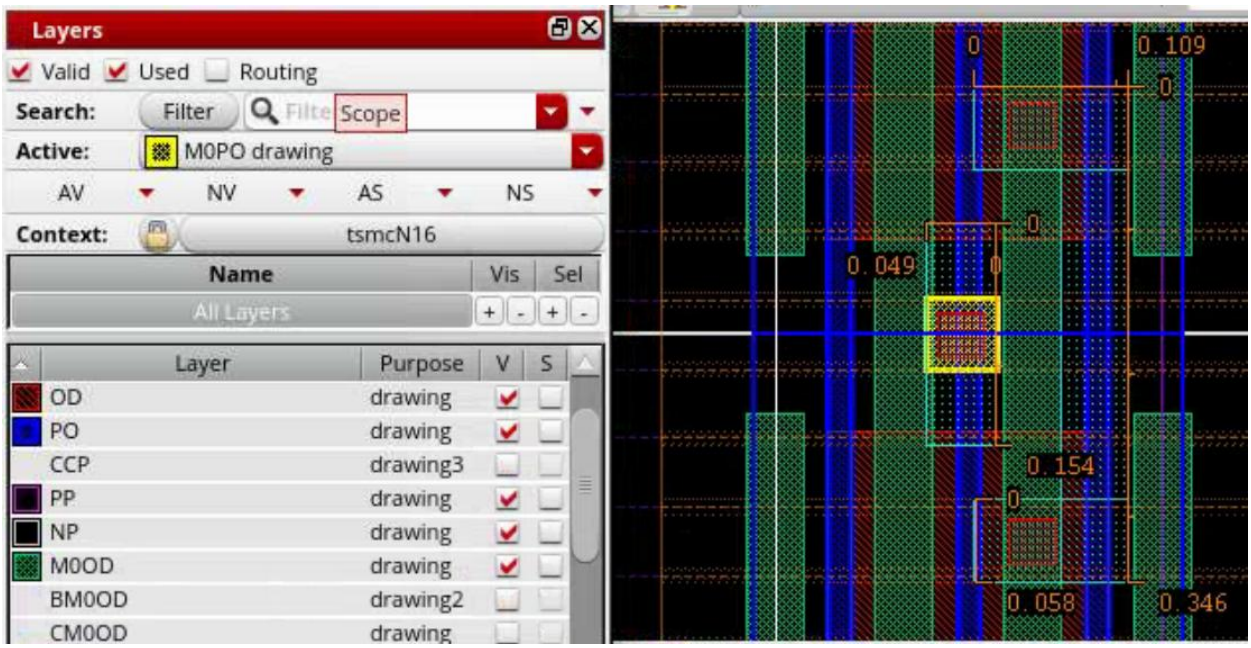
Lastly Add your **M1 drawing** or **Metal1** layers to create your metal connections. The VDD and VSS metal layer must be the same size and shape as each respective CCP rectangle.



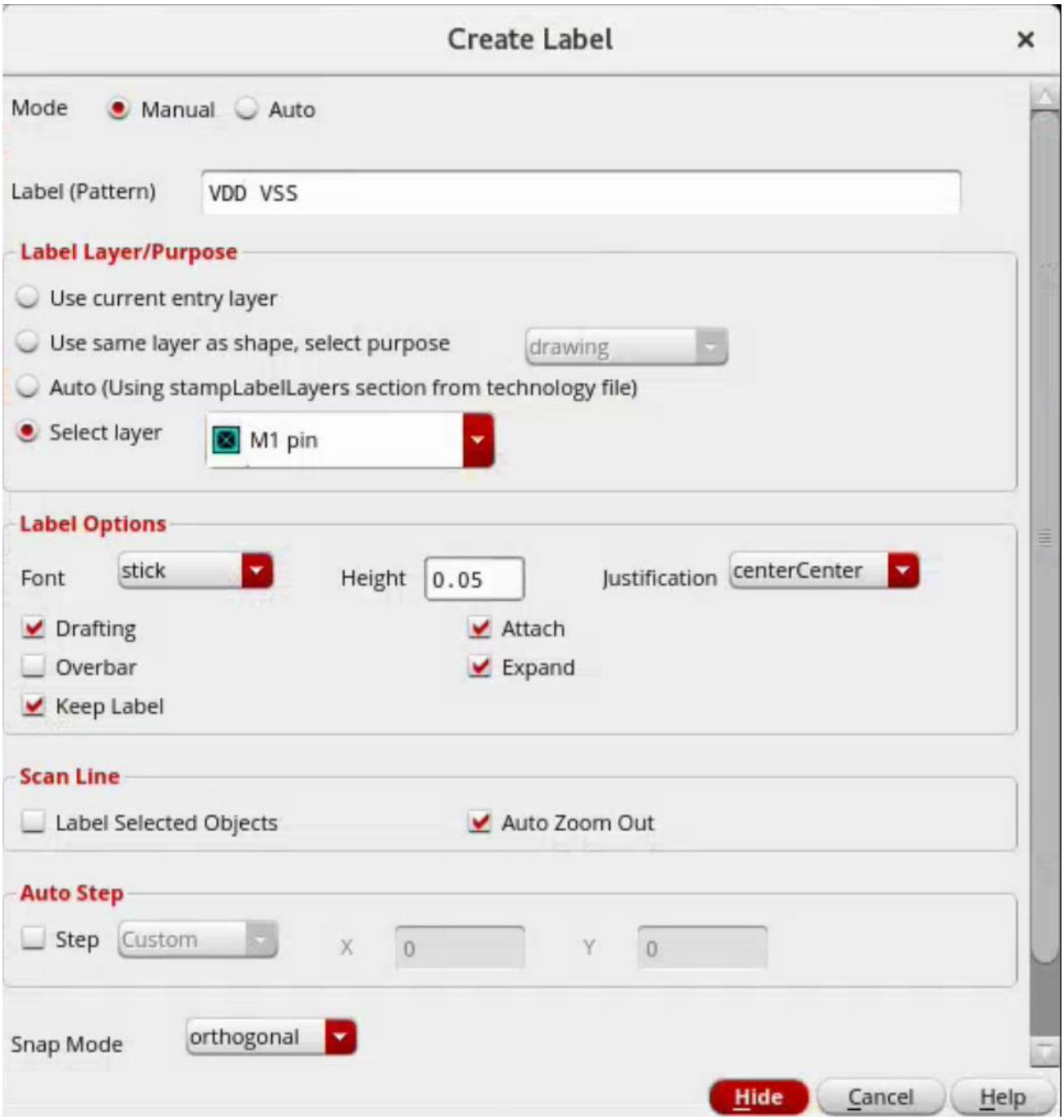


Your next Metal1 layers are ultimately up to you. They need to be wide enough to pass DRC and sperate metal1 layers must be far enough from each other to not cause DRC problems either. For more complicated cells it is up to you on how to route.

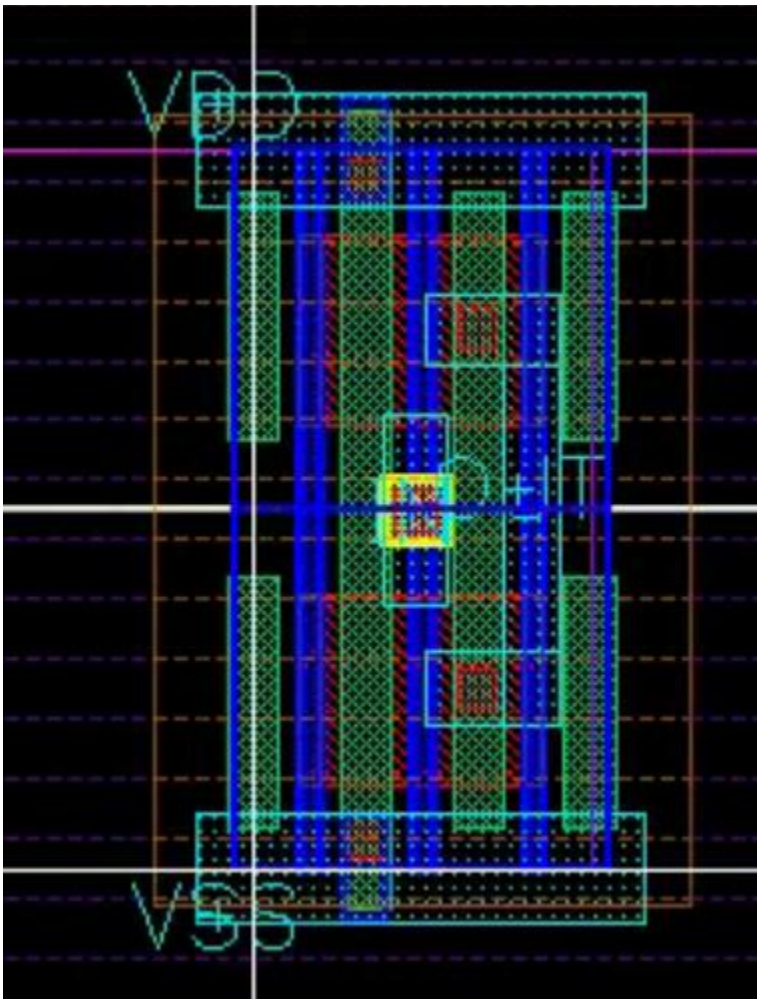




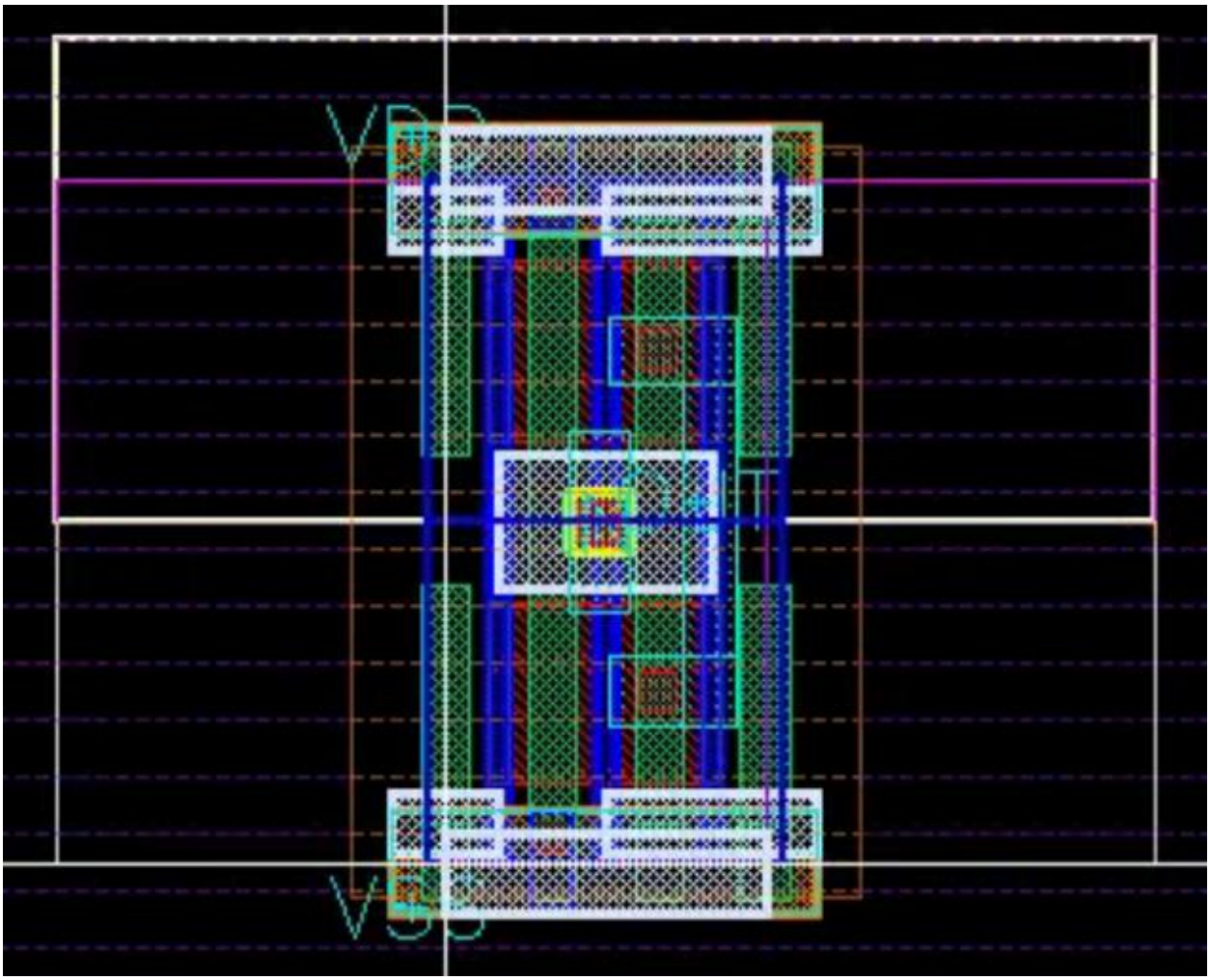
Lastly add your Metal1 Pins for **VDD** and **VSS.** Use **M1 pin layer**. **Create -> Label**



Place the labels on the **M1 drawing rect**. Make sure all layers are selectable. Does this for **IN** and **OUT** as well.

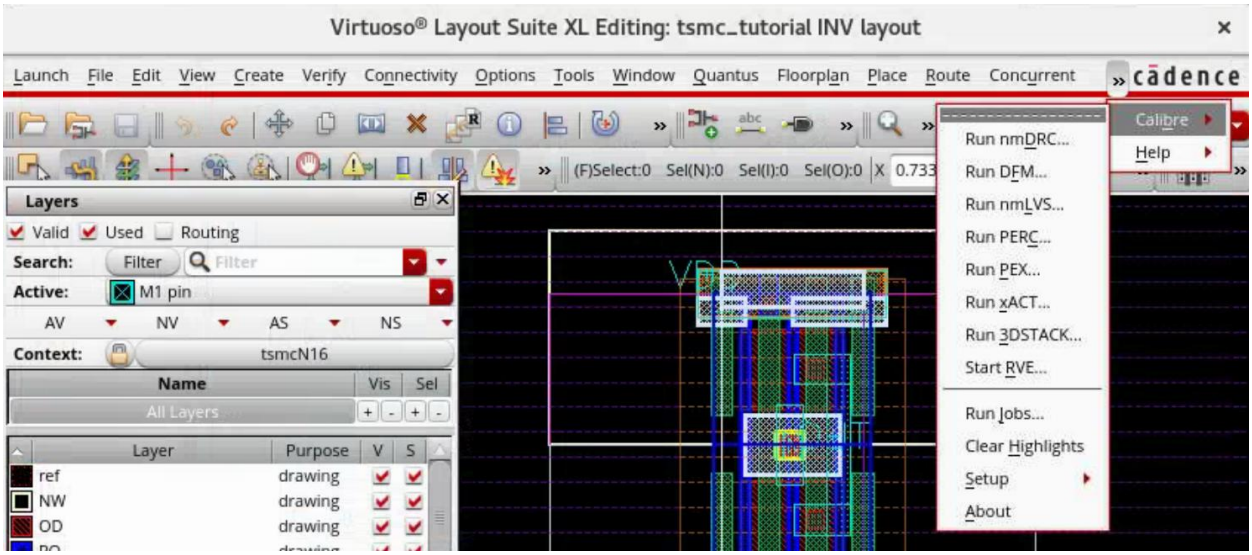


Full Cell.



Design Rule Check

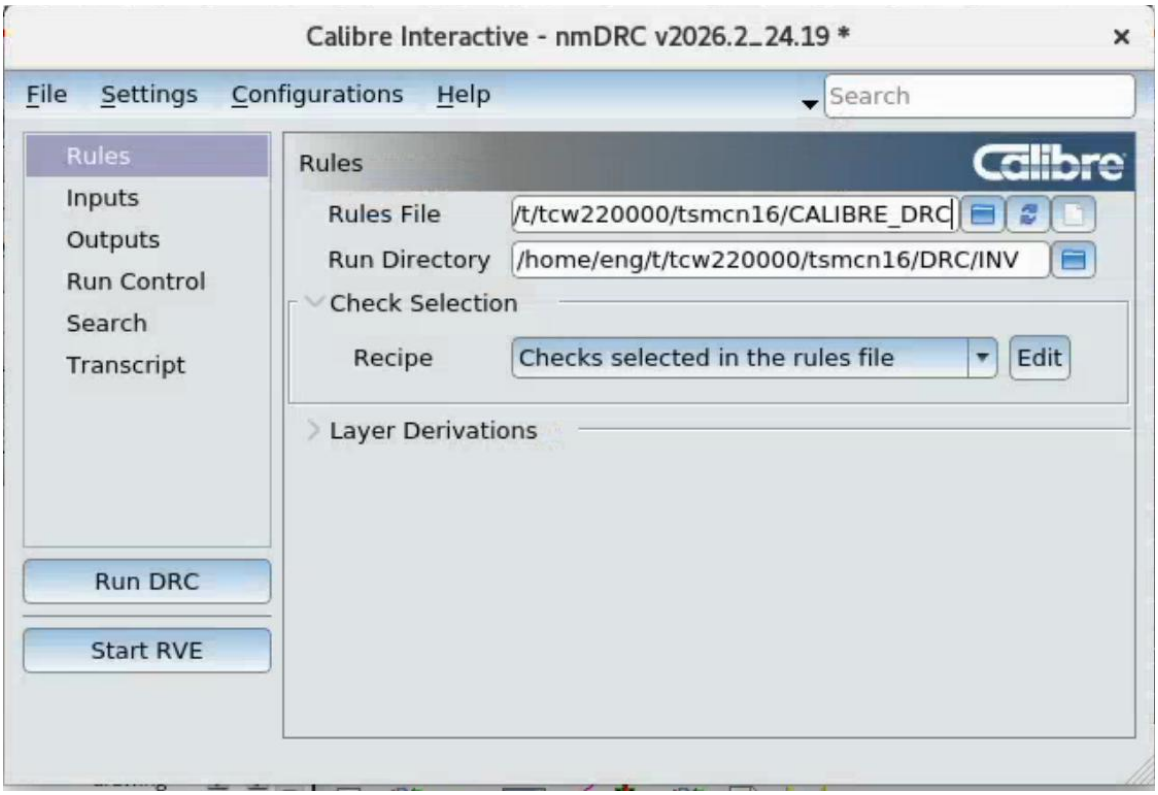
In the tool bar look for **Calibre -> Run nmDRC**



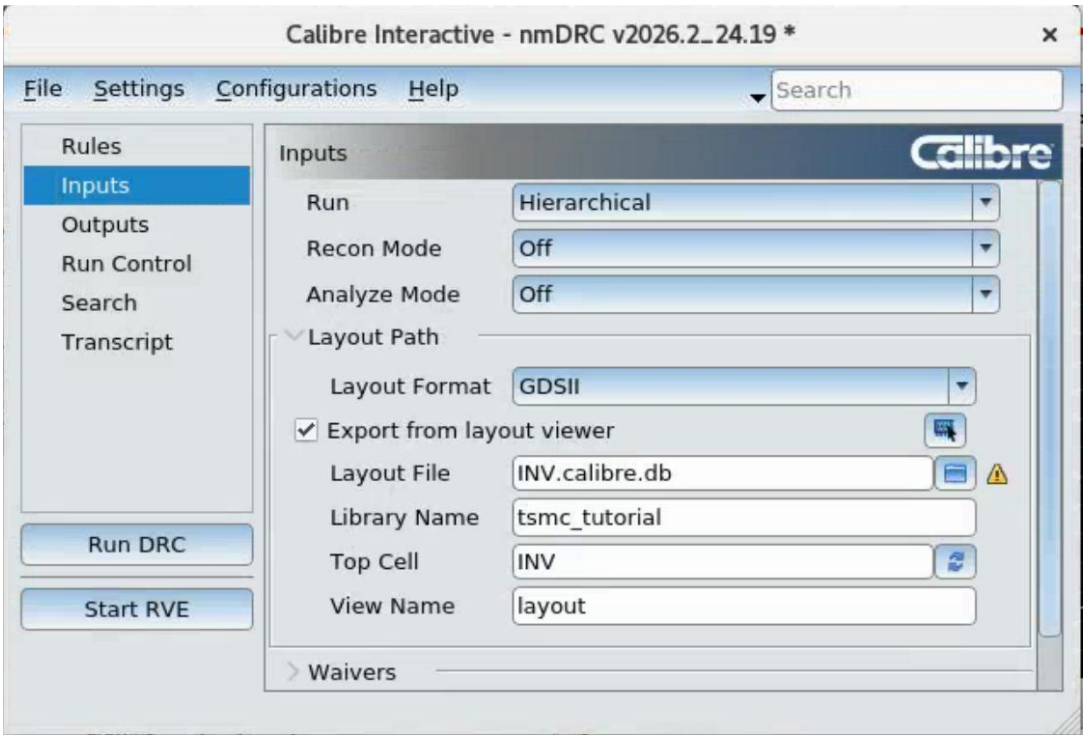
Click **Cancel** on the Load Runset Files.

Click the folder icon next **Rules File** and find the **Calibre_DRC** file in your work directory.

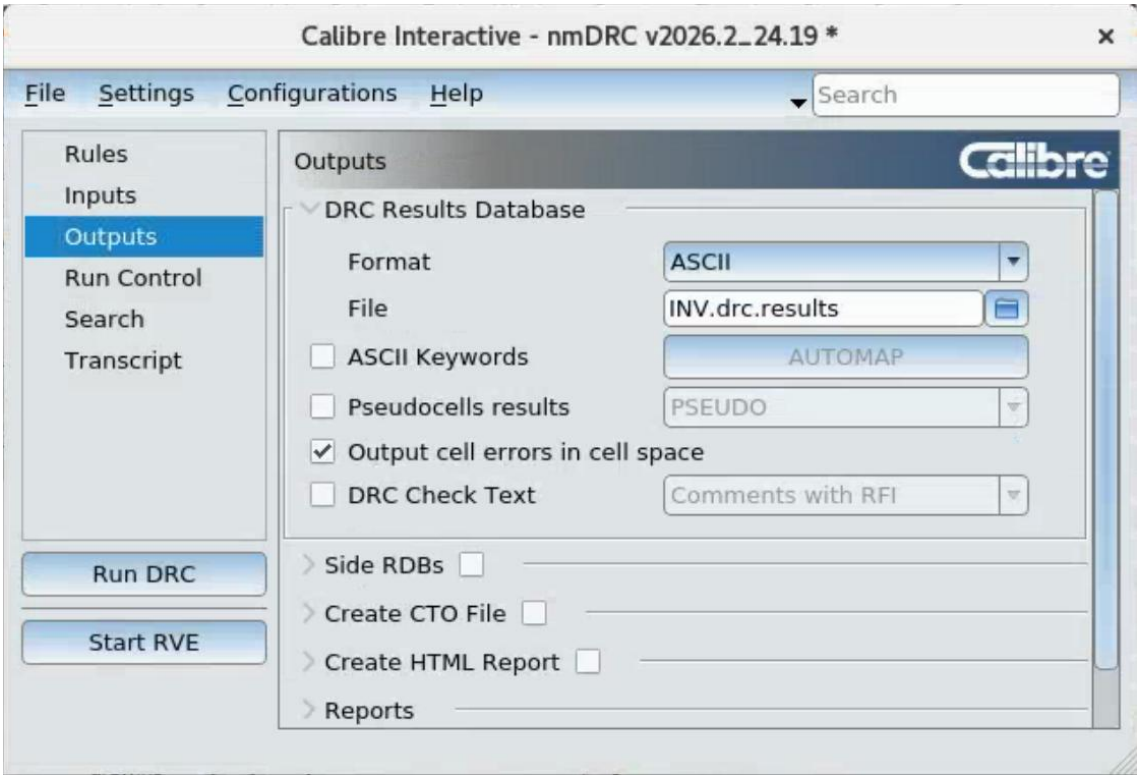
Click the folder icon next to the **Run Directory** and create a new folder **DRC** and then another folder inside DRC named **INV**. Then select that as your directory.



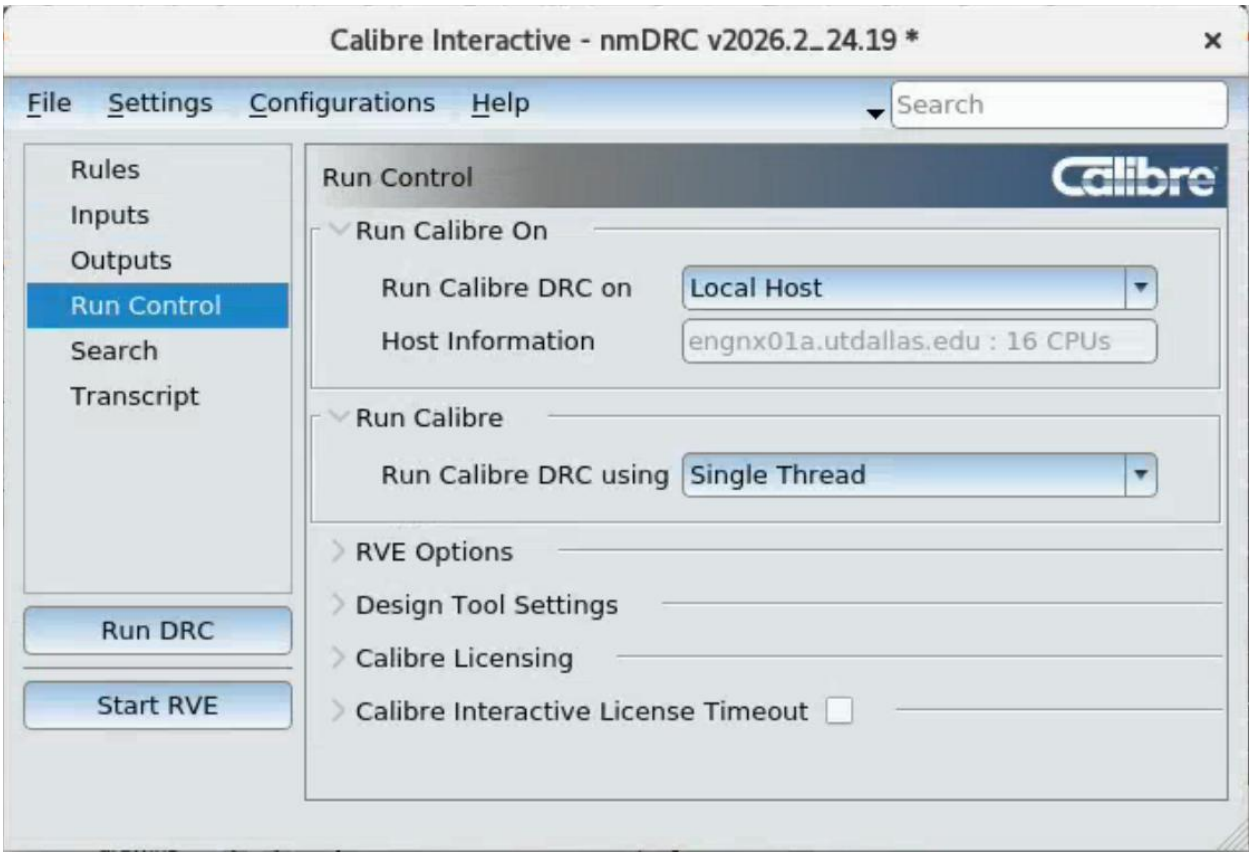
Inputs:



Outputs:

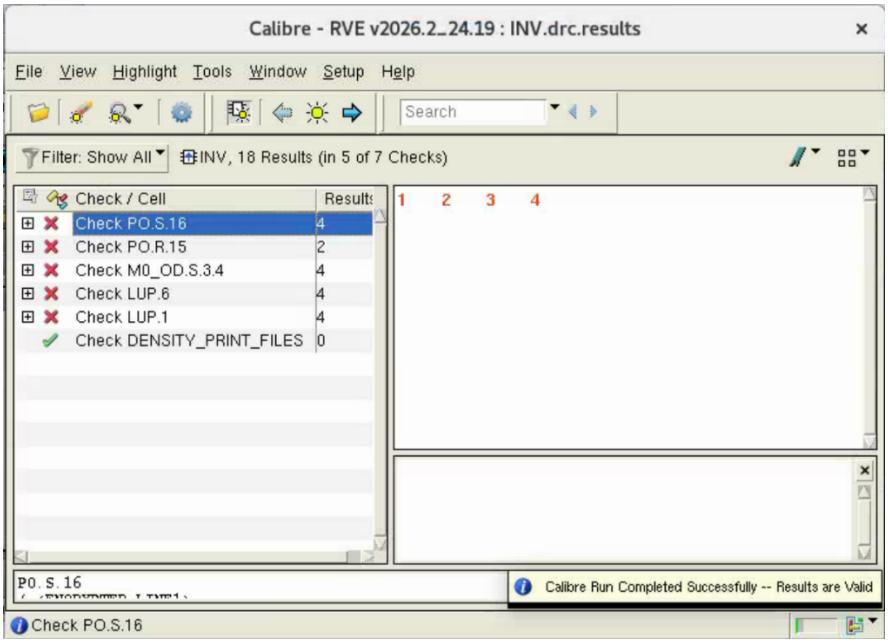


Run Control:



Then Click **Run DRC**. This will take a moment.

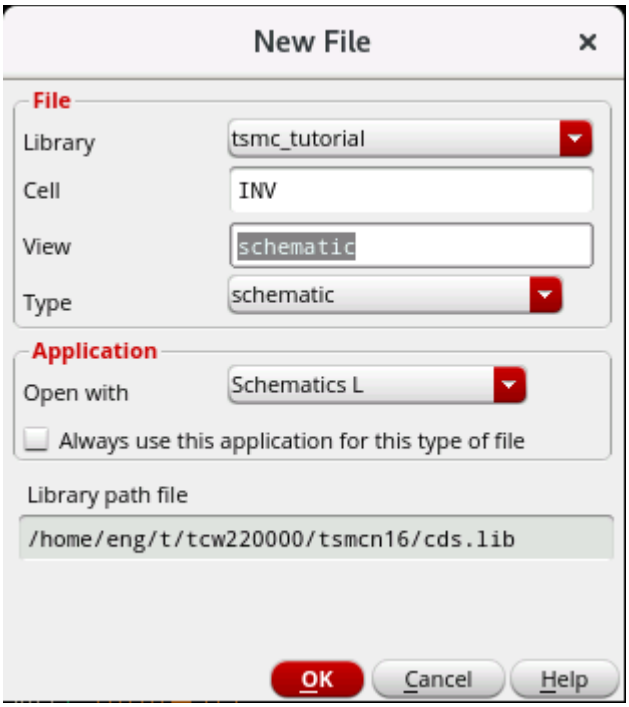
You should get these exact errors **PO.S.16, PO.R.15, M0_ODS.3.4, LUP.6, LUP.1**.



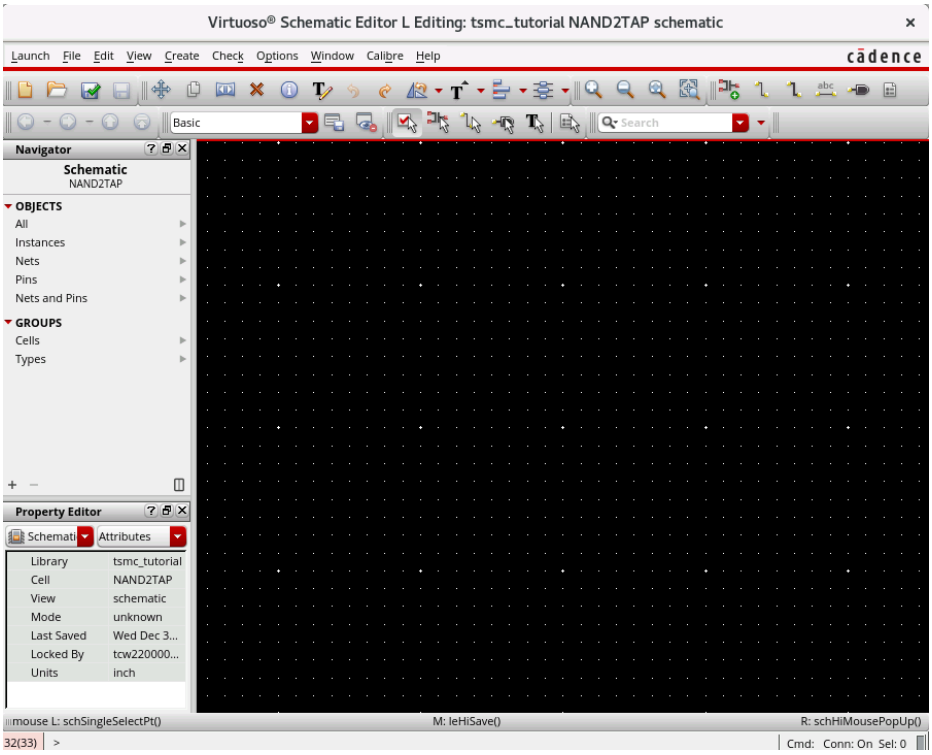
Layout vs Schematic

Create a new Cellview in the Library Manager.

File -> New -> Cellview



Click **OK**

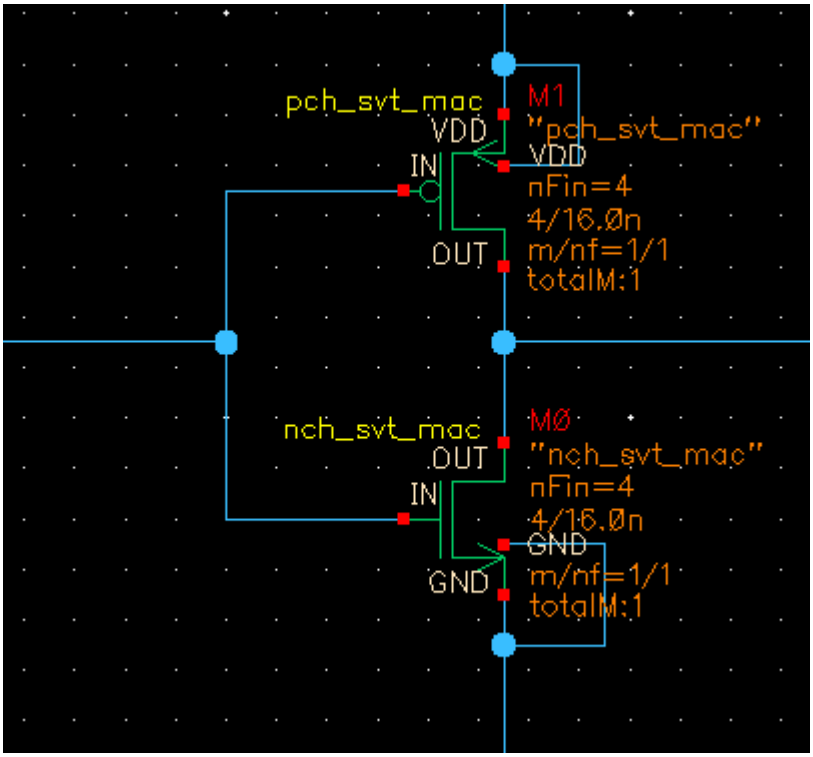


In the new window find the toolbar: **Create -> Instance**

Library: tsmcN16
Cell: nch_svt_mac
View: symbol
Number of Fingers: 1
Number of Fin: 2
width(M): 58n M
Length(M): 16n M
Source M1 Width(M): 38n M
Drain M1 Width (M): 38n M
Multiplier: 1

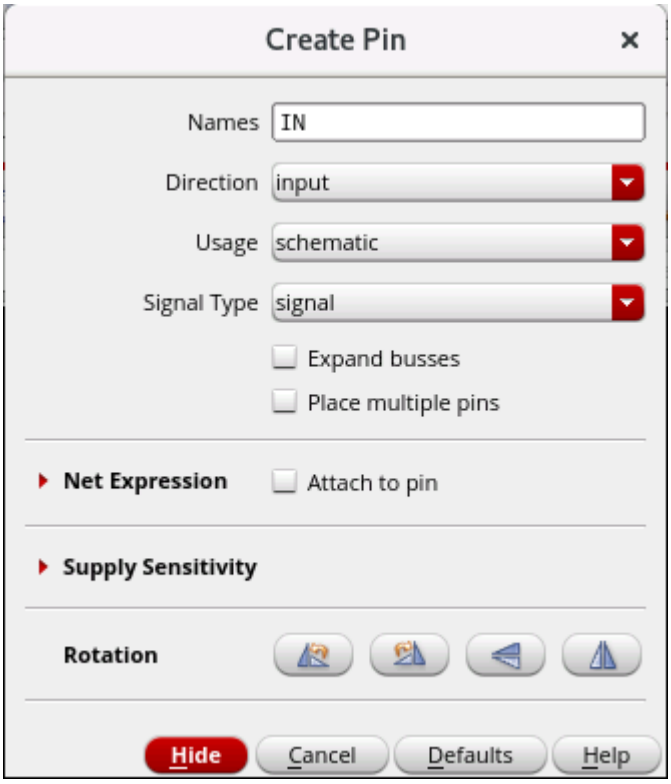
Similarly create a pfet device.

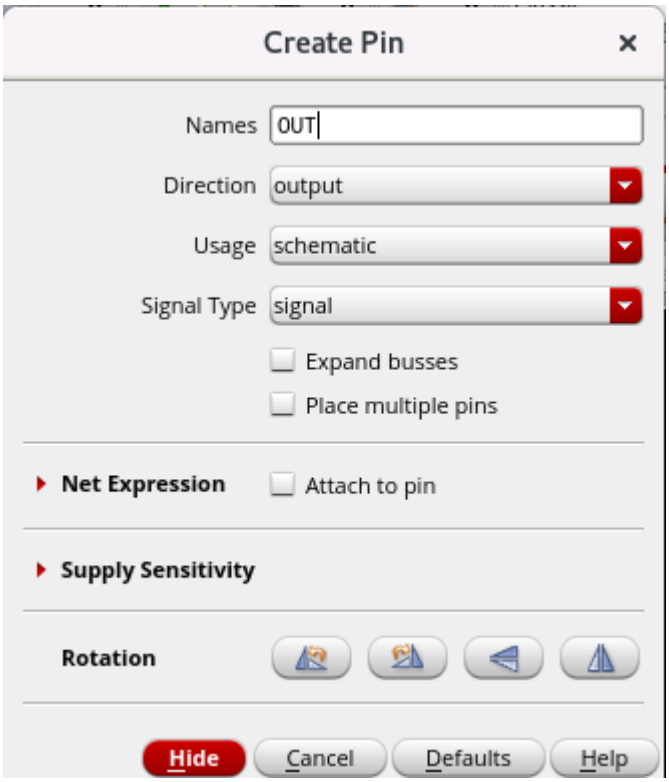
After placing the nfet and pfet press W and draw wires. When drawing wires do not group routing points together. If you have many wires running to one large blue dot then you will encounter an error when checking and saving later.

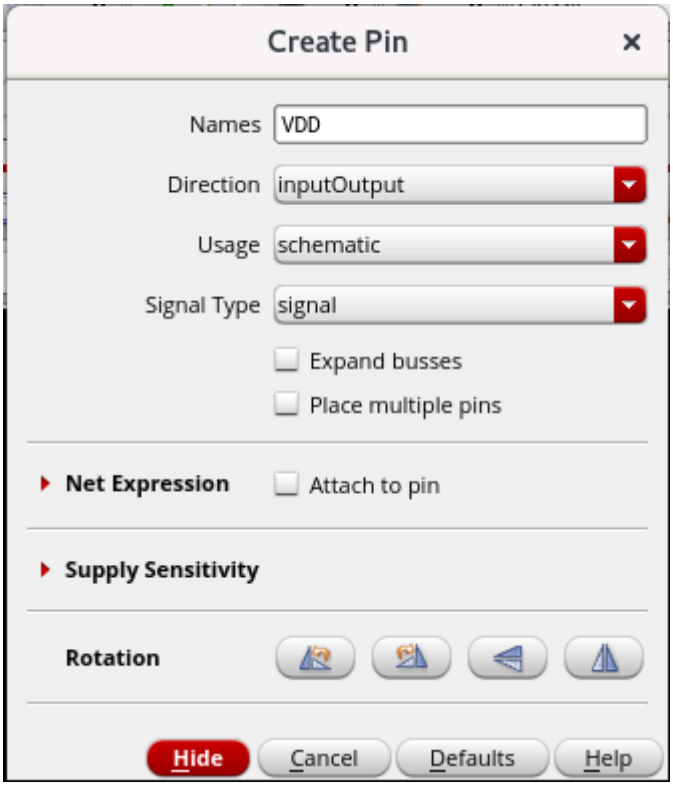


Create Input, Output, and VDD/GND Pins.

Create -> Pin





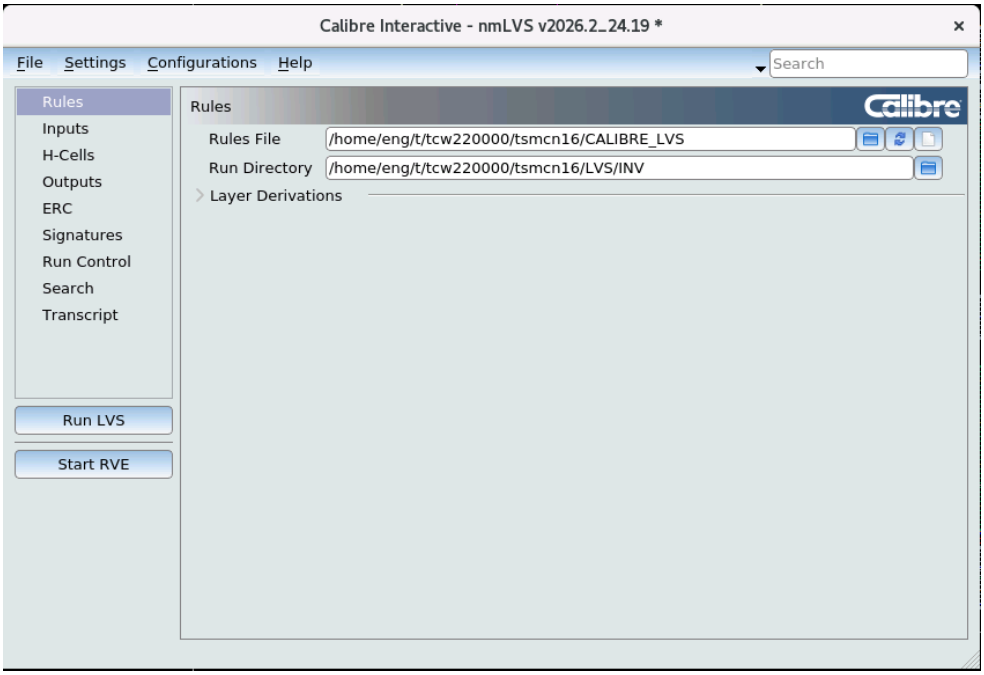


And the same for GND. For more complex cells with multiple Inputs extrapolate and create input A and input B, etc.

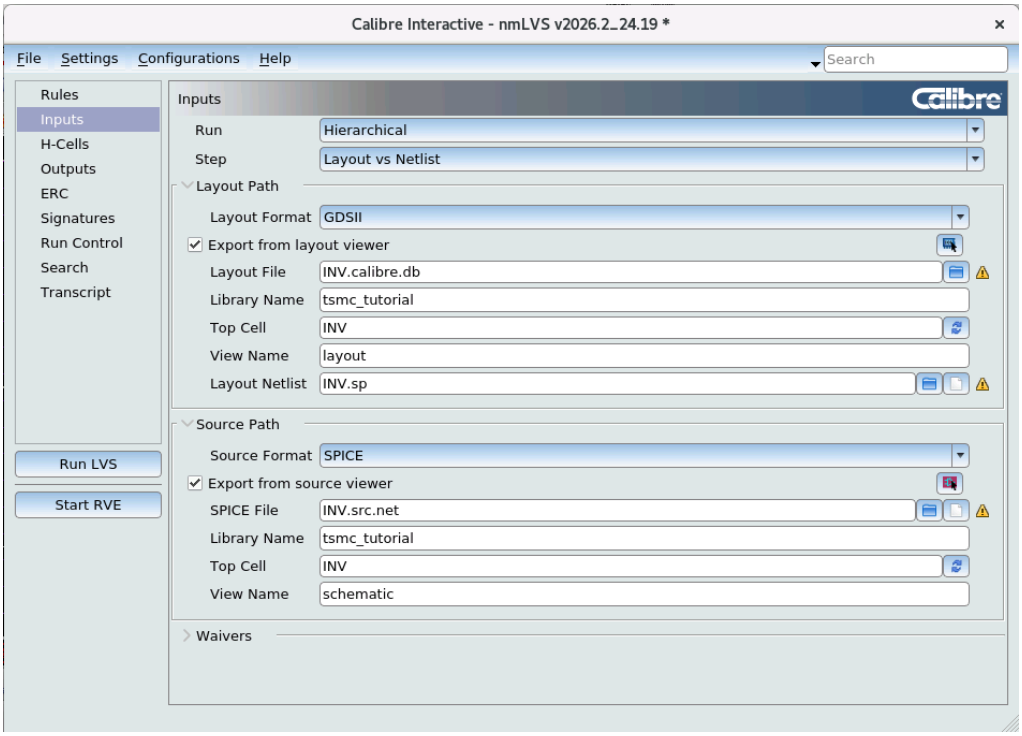
Check and Save the design.

In the the layout view, **Calibre -> run nmLVS**

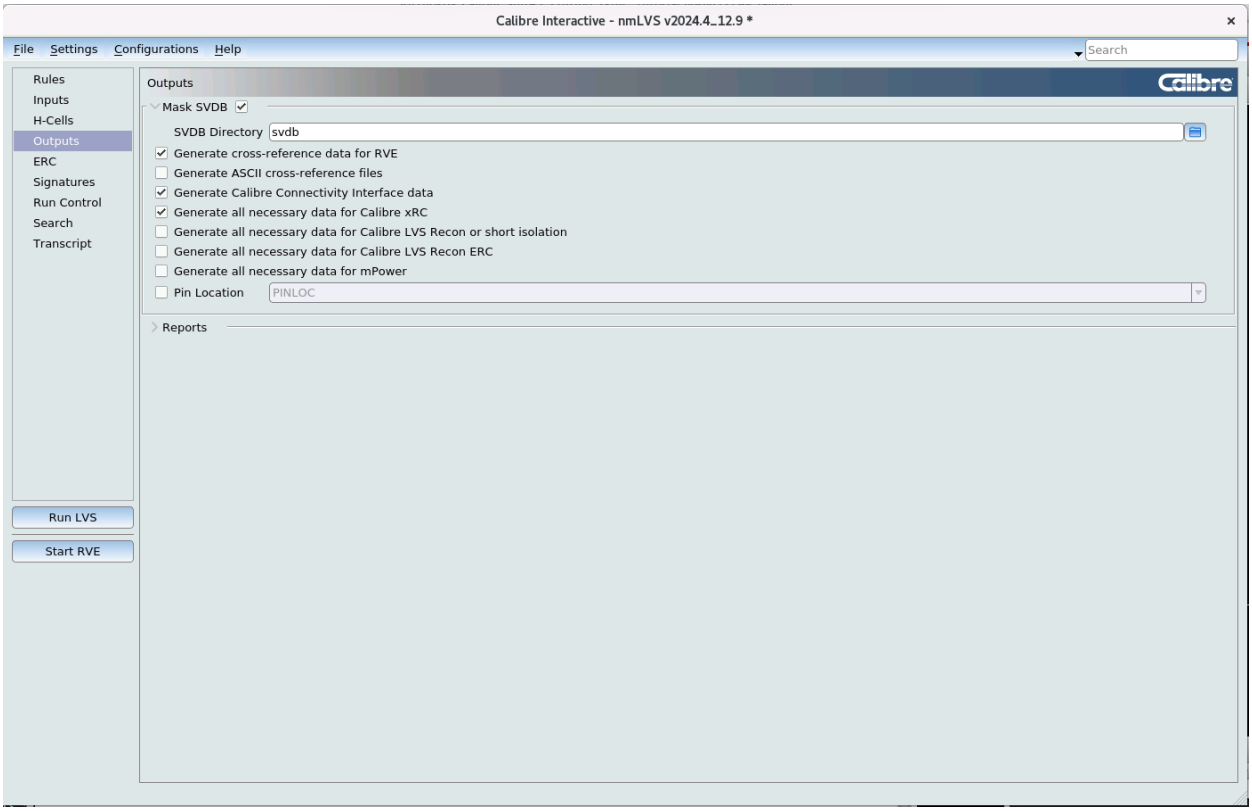
Close out of the initial window and find the CALIBRE_LVS file in Rules File using the folder icon. Using the folder icon next to **Run Directory** create an LVS folder and maintain it how you want. In this case there is an INV folder in the LVS folder to keep things organized.



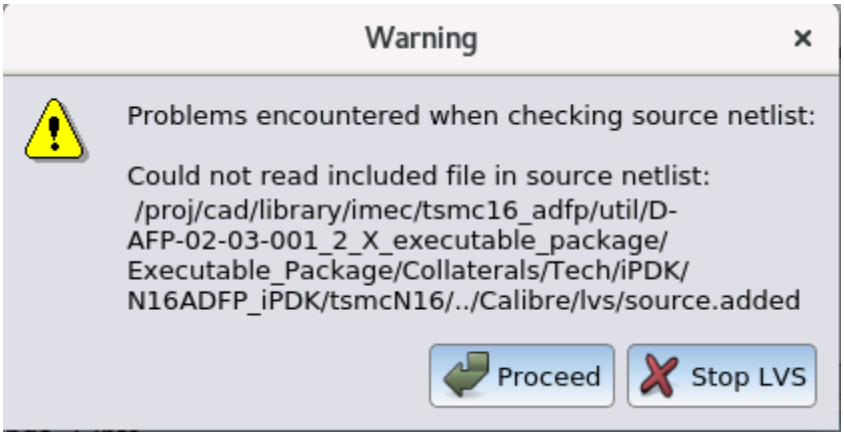
Chek your **Input Tab**



Check your **Output Tab**



If this is the first time running LVS for a cell. You might encounter an error:



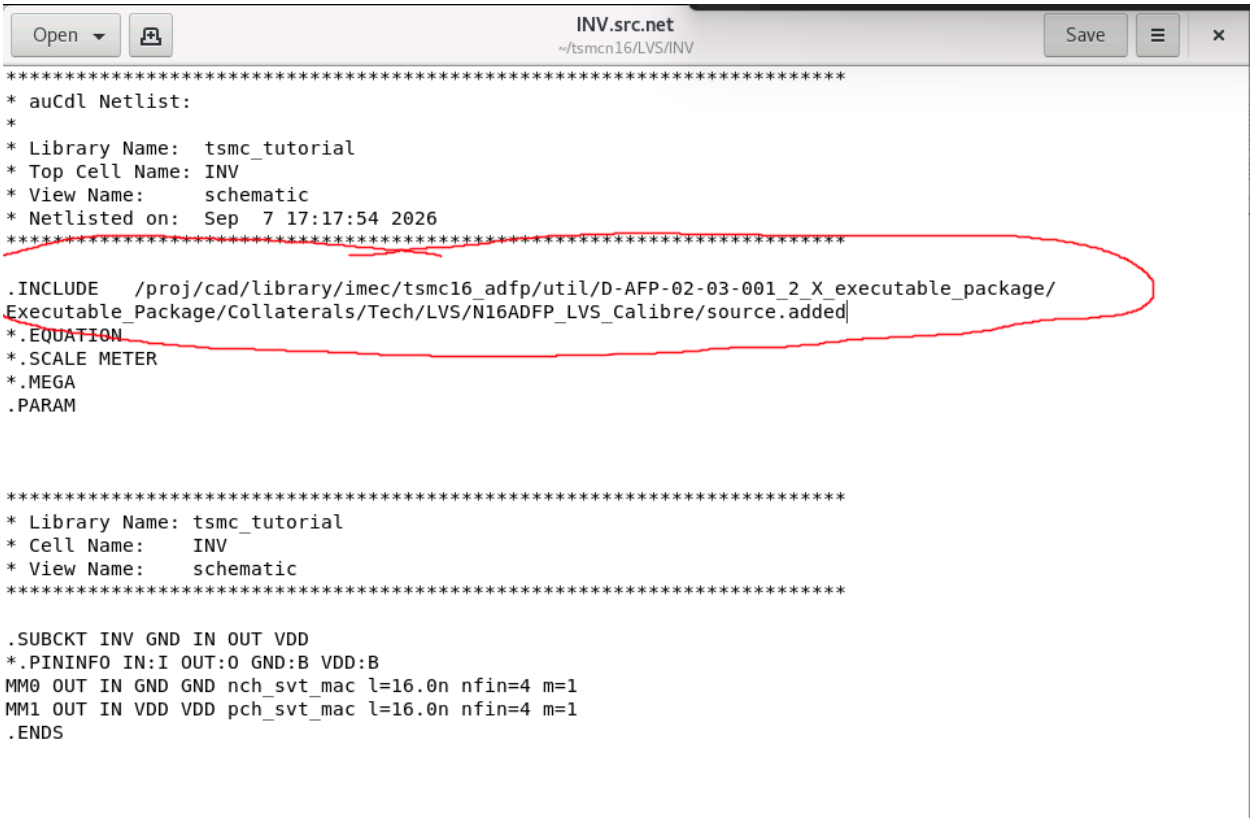
This is due to the autogenerated CELLNAME.src.net file directing the wrong INCLUDE package.

If you do not encounter this error, then the issue has been fixed, and you may skip to the TAPCELL section of this document on page 43.

Click **Stop LVS**

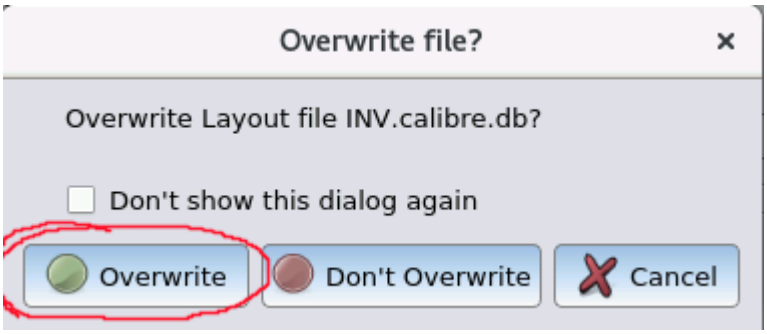
Navigate to your LVS folder and the cell folder you created. Find CELLNAME.src.net text file. Find the line: **.INCLUDE** and replace it with:

/proj/cad/library/imec/tsmc16_adfp/util/D-AFP-02-03-001_2_X_executable_package/Executable_Package/Collaterals/Tech/LVS/N16ADFP_LVS_Calibre/source.added

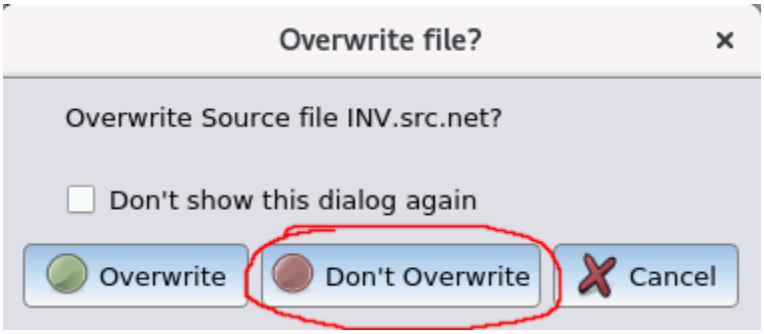


Rerun nmLVS.

Overwrite Layout file CELLNAME.calibre.db.

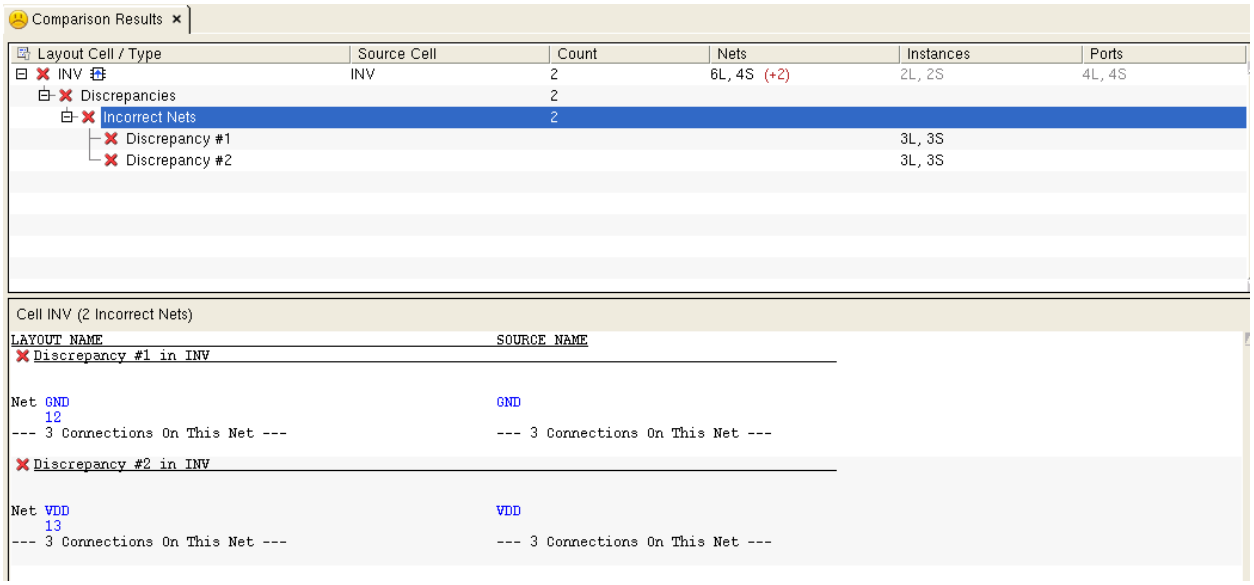


Do not Overwrite Source file CELLNAME.src.net



Note: If you make changes to the Schematic after this step you will have to Overwrite the CELLNAME.src.net and change the **INCLUDE** again.

You should run into an error.

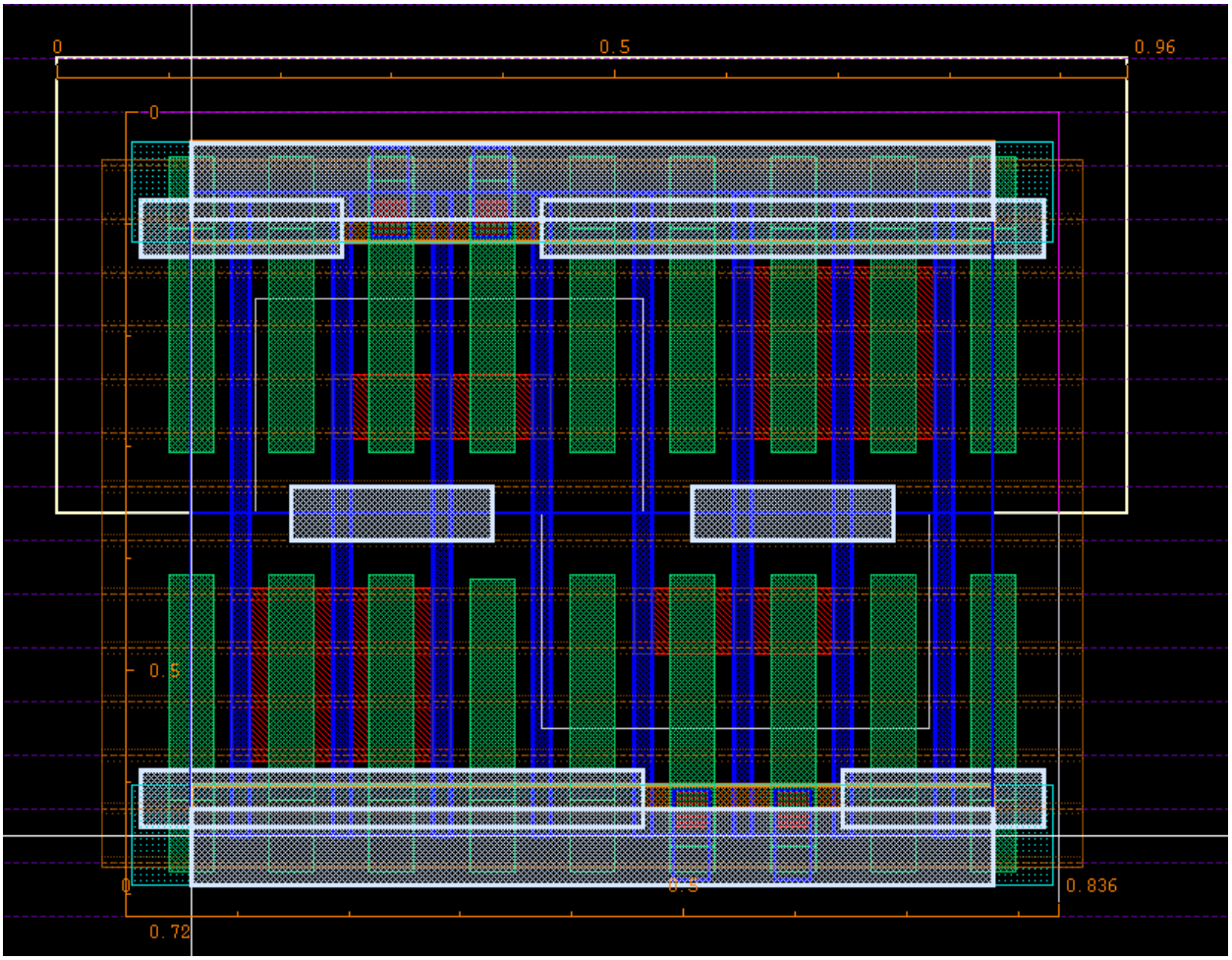


LVS is detecting 3 connections on the VDD and GND nets where it only expects 2. This is because we do not have TAPCELL applied to the cell to create the proper SOURCE-BULK connection.

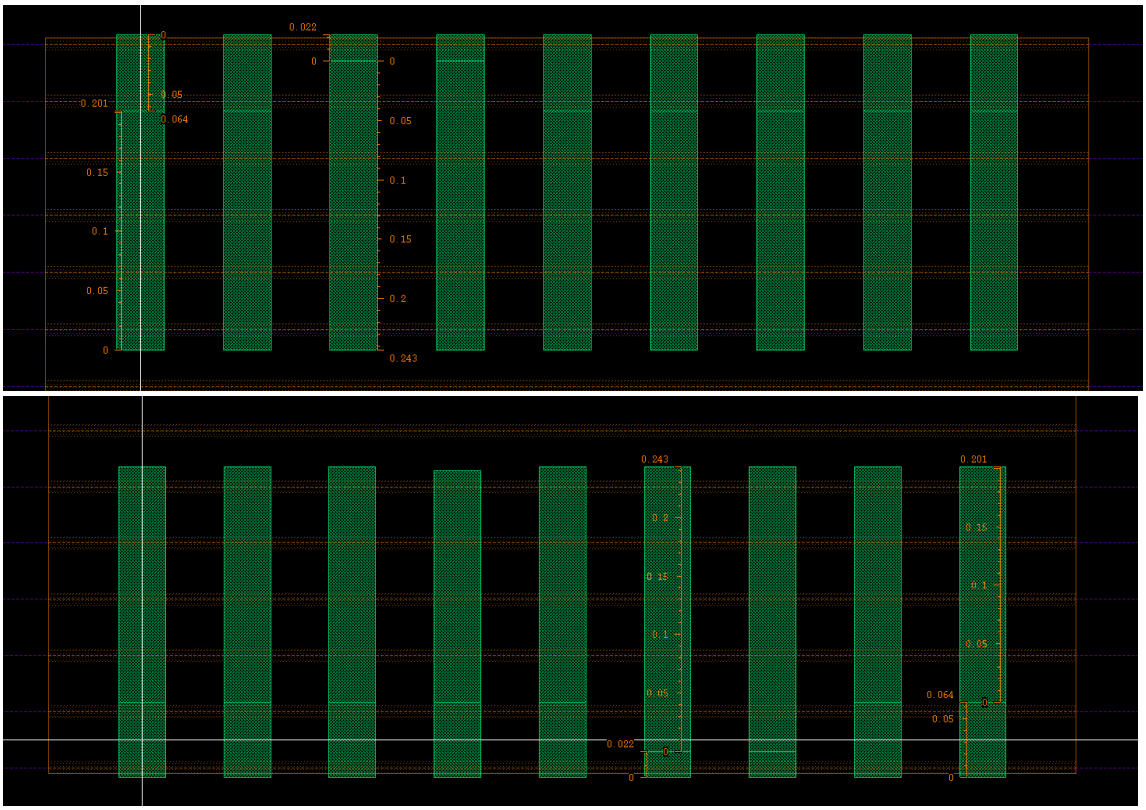
TAPCELL

Check the tsmcN16 Library for **TAPCELLBWP16P90**. If the cell does not exist, then you will have to create it yourself.

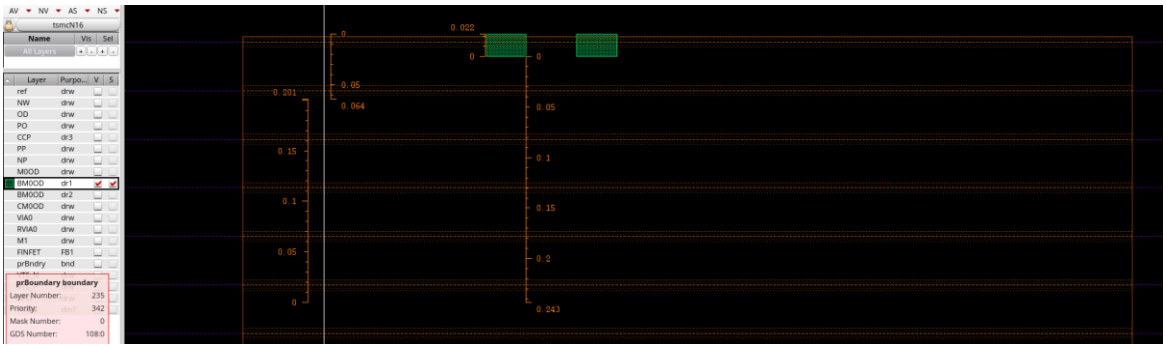
Below are images to help you create the TAPCELL. The same rules apply as the INV however you will notice the PP drw and NP drw are drawn rather interestingly.

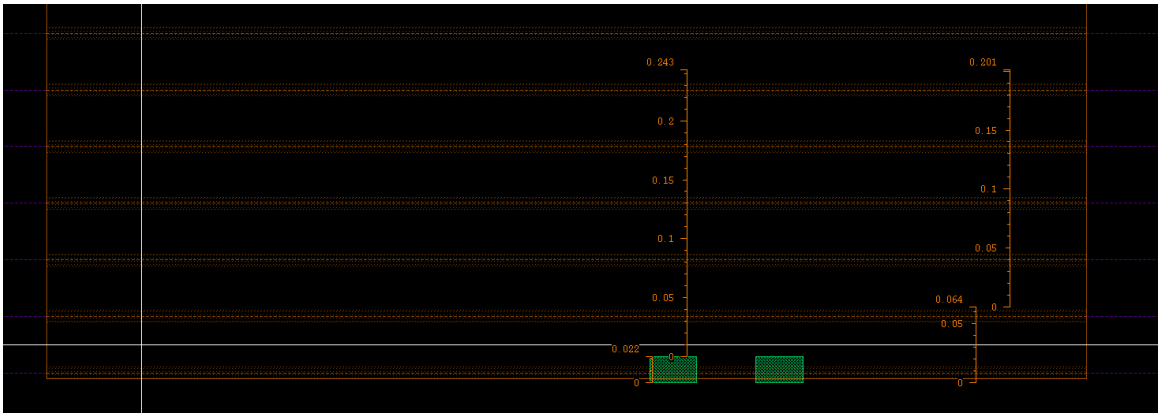


This cell is 7 Polys long and same height as before. Your Ref draw, NW draw, CCP dr3, CM0OD drw, Via0 drw, RVIA0 drw, M1 drw, Finfet FB1, prBndry, VTS_N drw, and VTS_P drw all are the same.

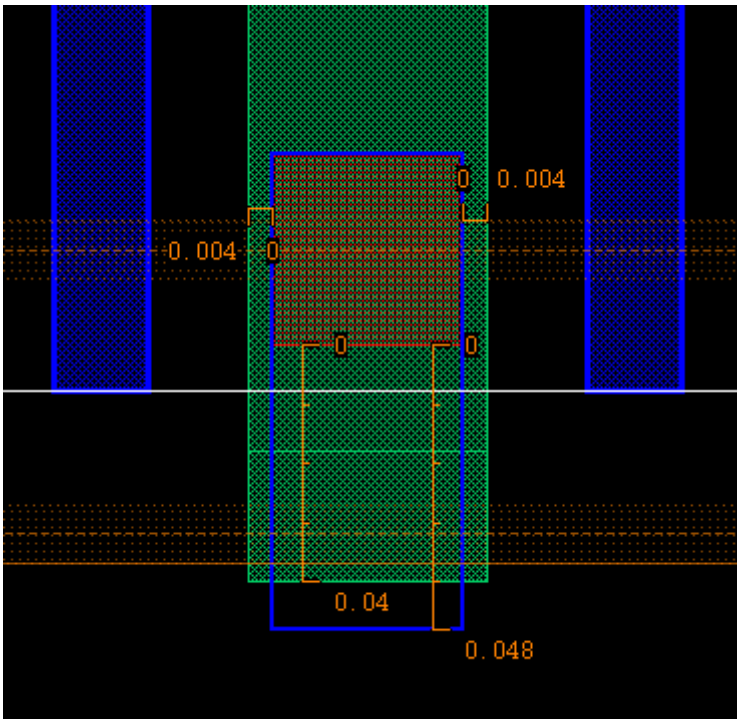


Notice the M0OD layer has a new layer **BM0OD dr1** which is the same idea as **BM0OD dr2**. The main M0OD layer is also physically separated.

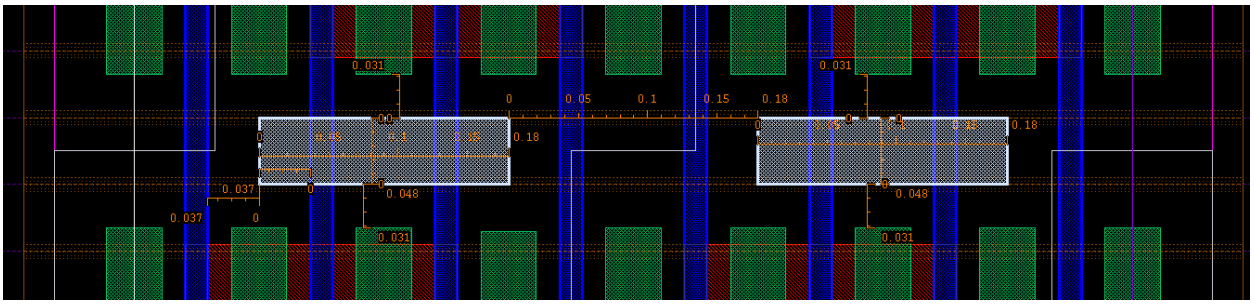




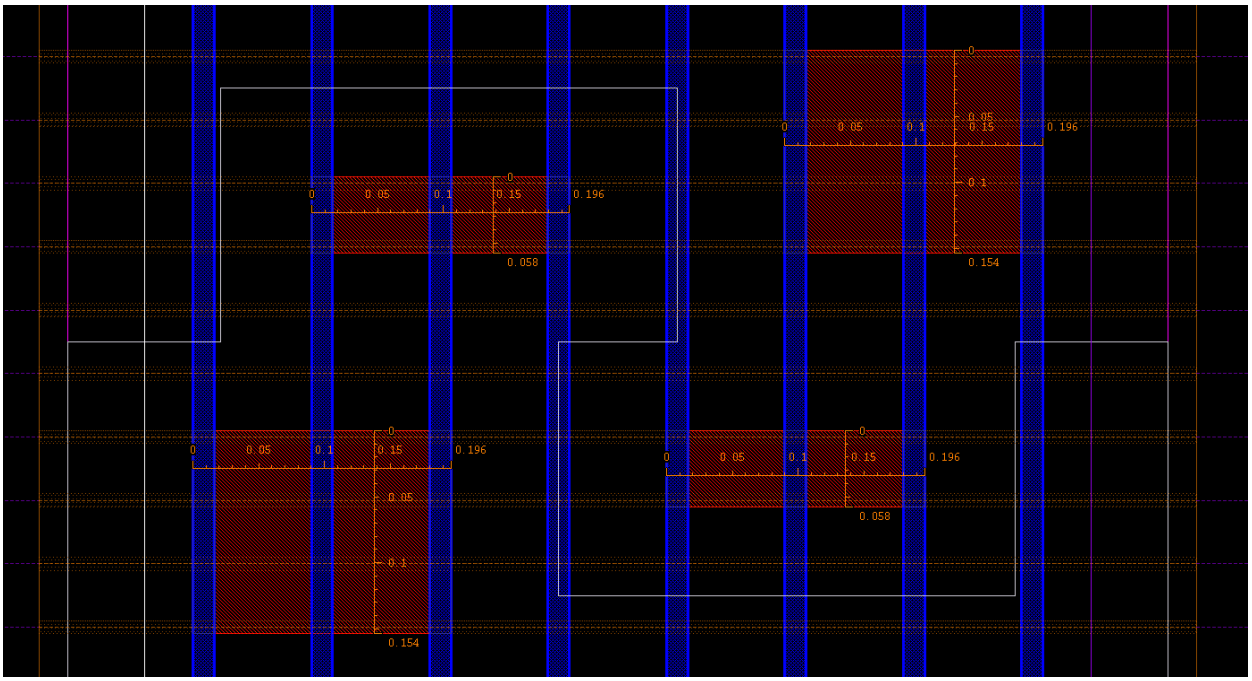
Align your Vias as such.

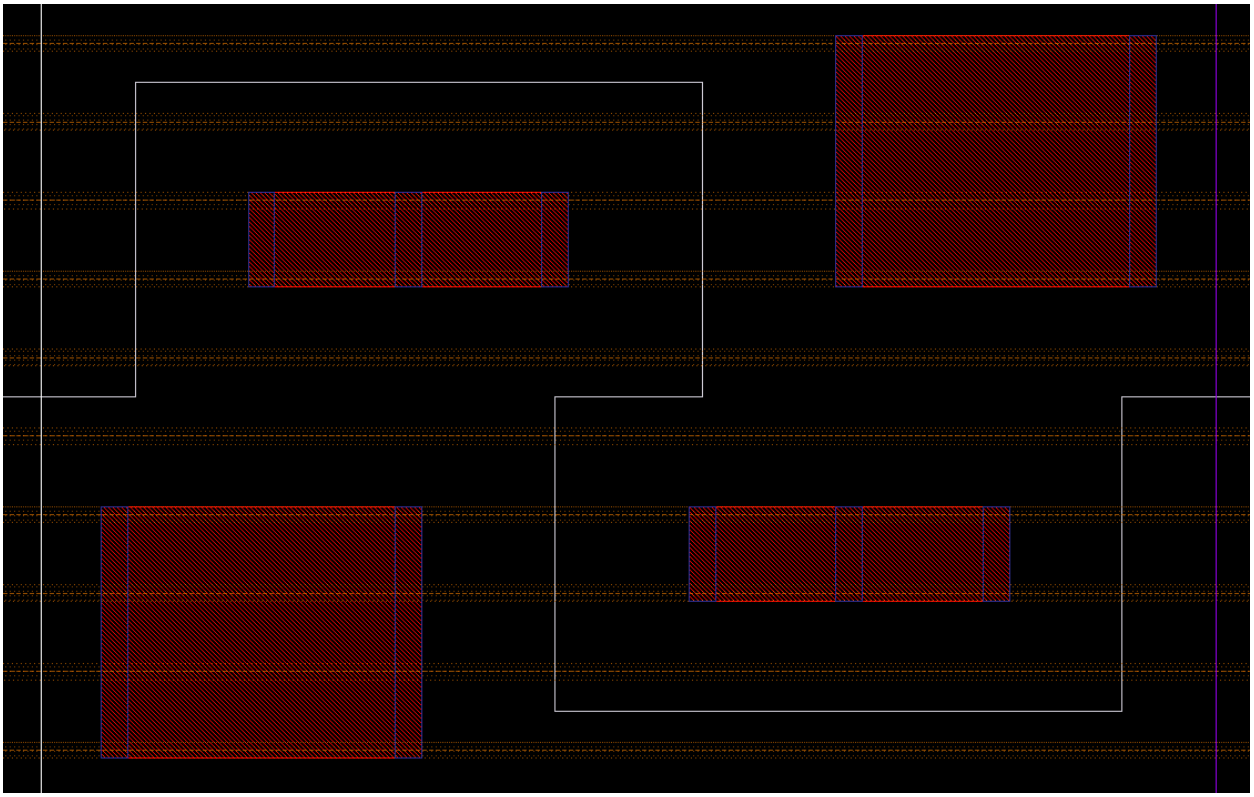


The CPO layer on the VDD and GND rails are the same but, in the middle, they are much smaller.



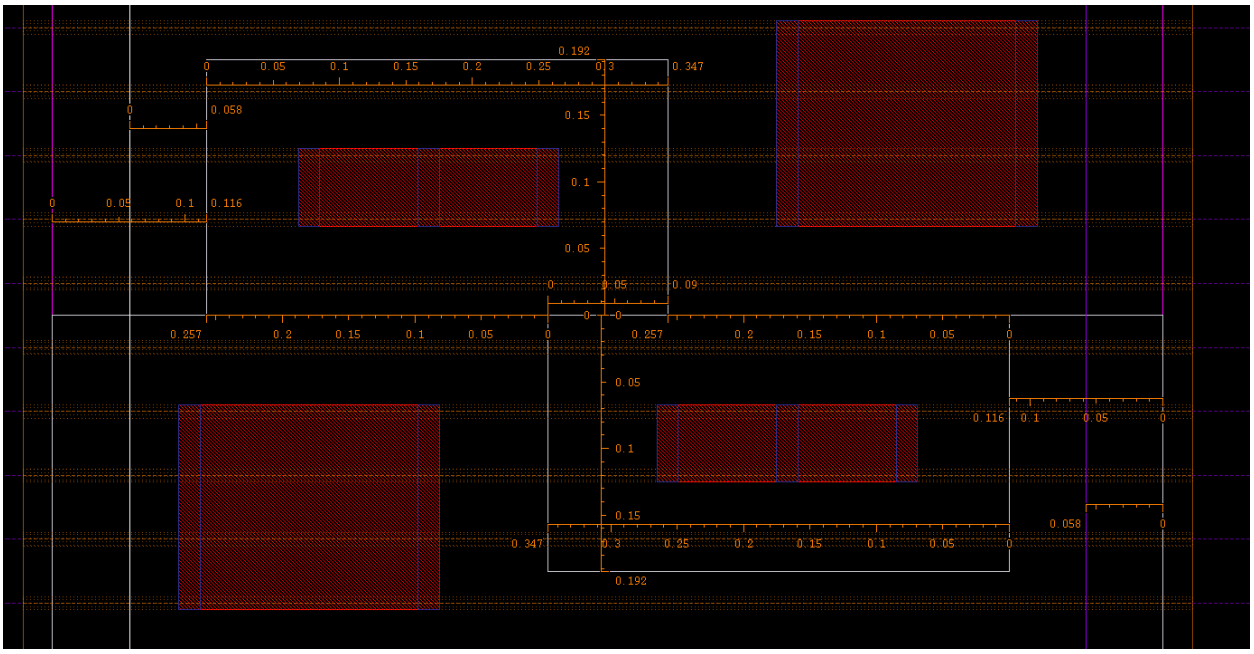
Draw OD layer as such.





Remember your PODE layer but notice that the smaller OD drawings have 3 PODE creating a 4Terminal device.

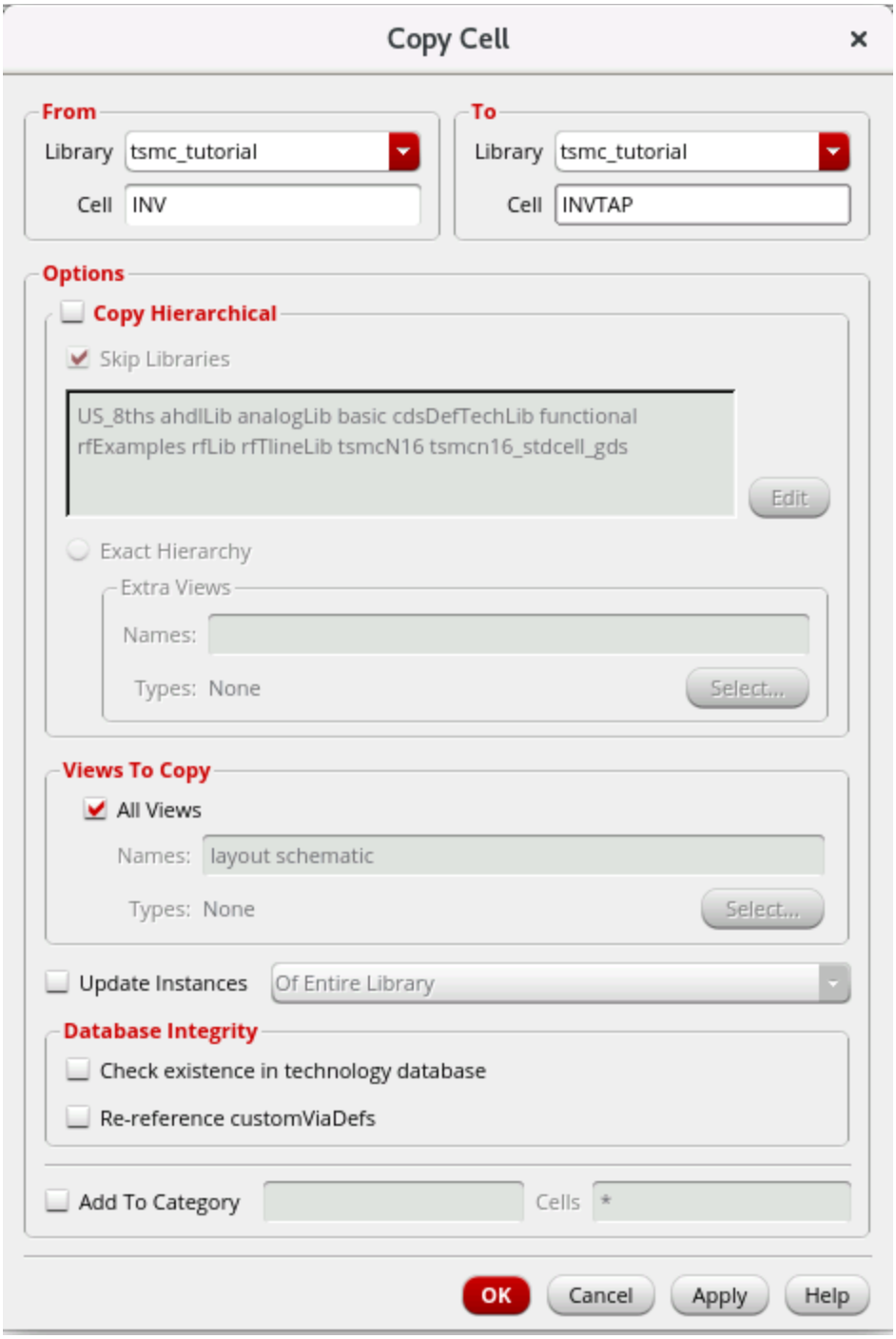
Lastly and most importantly, your PP and NP layers. Instead of drawing them vertically symmetric. You create pockets. This helps define your SOURCE-BULK connection.



LVS Continued

Create a Copy of your INV Cell.

Library Manger -> Under **Cell** Right Click **-> Copy ->** Rename the To section **INVTAP**



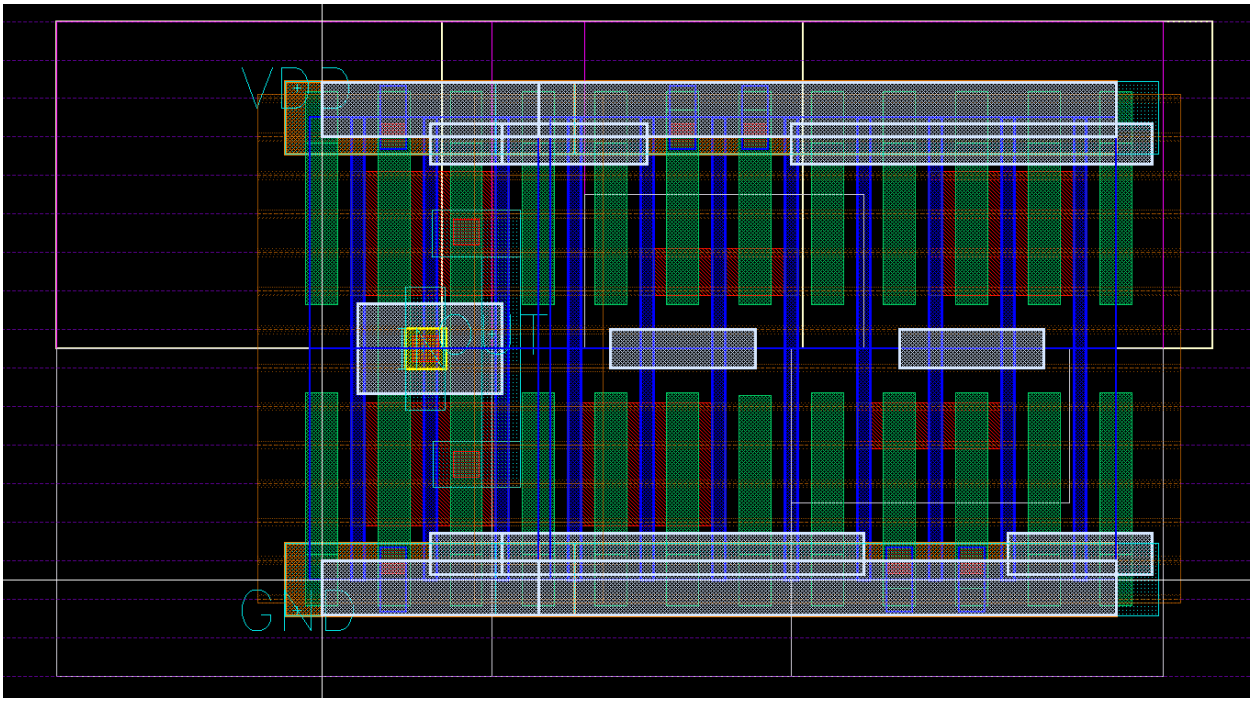
Click **OK**

Your **Schematic** and **Layout** should be copied under the new cell.

Open the **Layout** and Copy Paste the **TAPCELL** to the **INVTAP Cell**. Select all layers then hit **C** then drag your cursor to the layout window you wish to paste to.

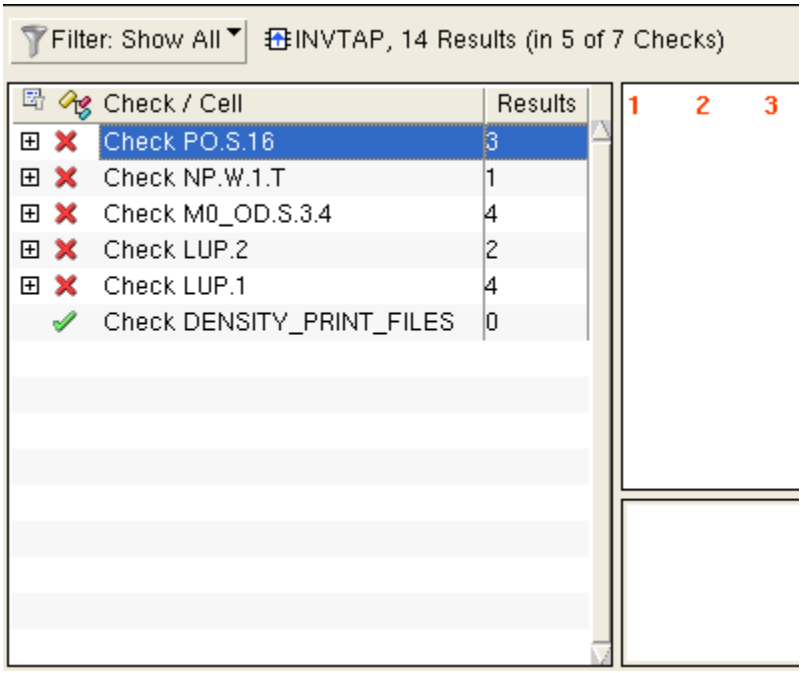
Check your PP and NP layers to make sure there are no overlapping issues. This can be done with the AV (ALL VIEW), NV (NO VIEW), AS (ALL SELECT), and NS (NO SELECT) buttons.

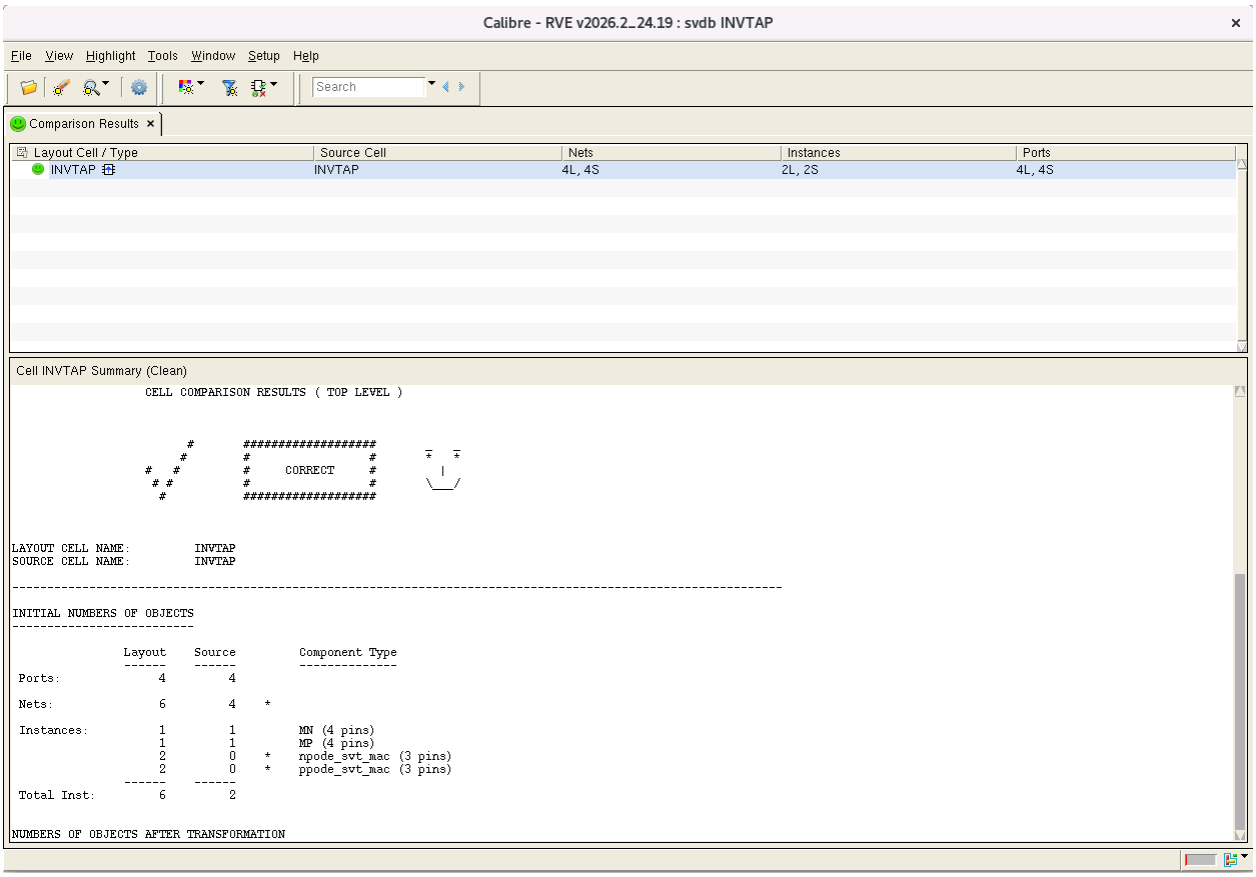
Make sure to put the Prboundary's flush with each other to properly abute the cells.



Your INVTAP Cell should look like this.

Perform DRC and LVS. DRC should lose the PO.R error and gain the NP.W.1.T error. Most importantly LVS should be clean.


Filter: Show All
INVTAP, 14 Results (in 5 of 7 Checks)
Check / Cell
Results
Check PO.S.16
3
Check NP.W.1.T
1
Check M0_OD.S.3.4
4
Check LUP.2
2
Check LUP.1
4
Check DENSITY_PRINT_FILES
0
1
2
3


Calibre - RVE v2026.2_24.19 : svdb INVTAP
File View Highlight Tools Window Setup Help
Search
Comparison Results
Layout Cell / Type
Source Cell
Nets
Instances
Ports
INVTAP
INVTAP
4L, 4S
2L, 2S
4L, 4S
Cell INVTAP Summary (Clean)
CELL COMPARISON RESULTS (TOP LEVEL)
CORRECT
LAYOUT CELL NAME: INVTAP
SOURCE CELL NAME: INVTAP
INITIAL NUMBERS OF OBJECTS
Layout Source Component Type
Ports: 4 4
Nets: 6 4 *
Instances: 1 1 MN (4 pins)
1 1 MP (4 pins)
2 0 * npode_svt_mac (3 pins)
2 0 * ppode_svt_mac (3 pins)
Total Inst: 6 2
NUMBERS OF OBJECTS AFTER TRANSFORMATION

PEX

Download ***pex.py*** from Canvas.

Place it in your ***tsmcn16*** directory.

Run:

Pyton3.11 pex.py --cell CELLNAME --cell-dir DRC/CELLNAME --lvs-dir LVS/CELLNAME --pex-dir PEX/CELLNAME

Example

```
{engnx03a:~/tsmcn16} python3.11 pex.py     --cell INVTAP     --cell-dir DRC/INV     --lvs-dir LVS/INVTAP     --pex-dir PEX/INV
```

If everything is correct you should see *PEX flow completed successfully*.

```
PEX flow completed successfully.
```

In the PEX folder, created by the script, find the *CELLNAME.cbest.pex.sp*.

PEX > CELLNAME > results > extracted_netlists

Open the *.sp* file and remove the numbers next to your I/Os, GND, VDD, and under TRANSISTOR CARDS, SUBCIRCUIT CALLS, and RESISTOR AND CAP/DIODE CARDS. This is important for primelib and for HSPICE now. The picture below is not accurate to the final output just an example.

Before

```
*       TRANSISTOR CARDS
*
*
MM0     OUT#12  IN#8    GND#1   GND     nch_svt_mac     L=1.6e-08       nfin=4
MM1     OUT#9   IN#7    VDD#1   VDD     pch_svt_mac     L=1.6e-08       nfin=4
*
```

After

```
*       TRANSISTOR CARDS
*
*
MM0     OUT#12  IN      GND#1   GND     nch_svt_mac     L=1.6e-08       nfin=4
MM1     OUT#9   IN      VDD#1   VDD     pch_svt_mac     L=1.6e-08       nfin=4
*
```

Save and close.

HSPICE

Make a new Directory for HSPICE Simulation in your work directory
mkdir ~/tsmcn16/spice
cd ~/tsmcn16/spice

Make another directory for INV files
mkdir inv
cd inv

Copy the *INVTAP.cbest.pex.sp* (this means you should have PEX you tap cells for this) file from your pex/inv directory to the new hspice inv directory.

Create a hspice file:
gedit inv_setup.sp

---

* Transistor Models

.LIB '/proj/cad/library/imec/tsmc16_adfp/util/D-AFP-02-03-001_2_X_executable_package/Executable_Package/Collaterals/Tech/SPICE/N16ADFP_SPICE_MODEL/n16adfp_spice_model_v1d0.l' FF_svt

.LIB '/proj/cad/library/imec/tsmc16_adfp/util/D-AFP-02-03-001_2_X_executable_package/Executable_Package/Collaterals/Tech/SPICE/N16ADFP_SPICE_MODEL/n16adfp_spice_model_v1d0.l' Total_svt

.LIB '/proj/cad/library/imec/tsmc16_adfp/util/D-AFP-02-03-001_2_X_executable_package/Executable_Package/Collaterals/Tech/SPICE/N16ADFP_SPICE_MODEL/n16adfp_spice_model_v1d0.l' stat

.LIB '/proj/cad/library/imec/tsmc16_adfp/util/D-AFP-02-03-001_2_X_executable_package/Executable_Package/Collaterals/Tech/SPICE/N16ADFP_SPICE_MODEL/n16adfp_spice_model_v1d0_aux.l' FF_mpodesvt

.LIB '/proj/cad/library/imec/tsmc16_adfp/util/D-AFP-02-03-001_2_X_executable_package/Executable_Package/Collaterals/Tech/SPICE/N16ADFP_SPICE_MODEL/n16adfp_spice_model_v1d0_aux.l' Total_mpodesvt

* extracted cell

.include 'NAND2TAP.cbest.pex.sp'

.option POST=2 RUNLVL=5

X_DUT OUT GND VDD A B NAND2TAP

* Supplies and I/O Pulses. Make changes as needed

V_GND GND 0 0

V_VDD VDD GND 0.88

VA A GND pwl(0ps 0v 2000ps 0v 2012ps 0.8v 4000ps 0.8v)

VB B GND pwl(0ps 0v 1000ps 0v 1012ps 0.8v 2000ps 0.8v 2012ps 0v 3000ps 0v 3012ps 0.8v 4000ps 0.8v)

* Load

COUT OUT GND 2.0000000000000003e-14

* Transient analysiss

.TRAN 1e-12 5e-09

.end

Save the file and run HSPICE:
. /proj/cad/startup/profile.synopsys_2018
hspice INV_setup.sp

Job Concluded.

NOTE: Check you copy pasted everything correctly.

HINT: This setup file is not made for an INV. **YOU** need to figure out what to change to make it correct. PINS and PIN order between your setup and sp file are important.

Move to the Waveview Tutorial.